José Rizal University

NON-INVASIVE CHOLESTEROL SCREENING DEVICE USING HAND IMAGING AND ARTIFICIAL NEURAL NETWORKS

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Approval Sheet

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Chapter 1

INTRODUCTION

This chapter provides an overview of the study, presenting its background, problem statement objectives, and the proposed solution using non-invasive AI-based cholesterol monitoring.

1.1 Background of the Study

High cholesterol is a silent yet a severe health issue which significantly puts the vulnerability to cardiovascular diseases that cause the most cases of death in the world. Low density lipoprotein (LDL) cholesterol commonly known as the bad cholesterol has the potential of causing heart attacks, stroke and other life-threatening diseases. Monitoring cholesterol on a regular basis is very important in prevention but more people fail to undergo the tests because of the inconvenience and the pains that come with the conventional blood-based laboratory tests.

Testing the level of cholesterol involves currently an invasive procedure of extracting blood which is later analyzed in a clinical laboratory. Although this is effective it is usually time consuming, expensive, and not feasible at regular or preventive screening particularly in underserved or rural areas where accessibility to health care facilities is poor. The obstacles elevate the rate of undiagnosed and untreated cases that lead to unnecessary health conditions and escalated health expenses.

The rest recent development has explored the possibility of non-invasively monitoring cholesterol via artificial intelligence (AI) and smartphone based-imaging. One such system was the Choldet by (Umar et al., 2020) that was implemented using a camera of a smart phone to scan the palm of the user and measure the level of cholesterol using artificial neural networks. Another project was NICMS by Abiramee et al. (2024), which incorporates photodiode sensors and AI models to investigate light absorption through the use of skin. These were some good prospects but in real sense they had their flaws like the small size of samples, varied lighting, fewer clinical confirmations and inconsistency of results they provided due to camera differences or user miscalculation.

Two gaps created relative to that aim are to be filled by this project: a specific non-invasive cholesterol tracking hardware platform, a compact sensor box, which new users put their hand in one of the sensor compartments. The device records the amount of light absorbed and skin response by using photodiodes and optical sensors inside the device. The data is then run through an embedded AI model and projections on the levels of cholesterol are obtained.

The output data is transmitted in real time to a health plan secure mobile-based application where participants can see the history of their results, and get health suggestions. This system is user-friendly, portable and installable in clinics, and at home.

The project hopes to offer a set, repeatable procedure that is fully non-invasive and capable of tracking the level of cholesterol in the body without any use of the blood samples or visits to the lab. It promises to be a positive change in

the communities with low access to healthcare and facilitate early detection of cardiovascular risks, by enabling constant monitoring.

The project will be focused on creating a fixed, repeatable and, overall, non-invasive method of checking the cholesterol level without the use of blood samples or lab visits. It will enable the community with low healthcare access to have continuous and accessible monitoring as well as facilitating the early detection of cardiovascular risks. In particular, the study will examine the application of the AI-based hand imaging as an alternative to traditional methods of cholesterol monitoring in terms of its convenience, comfort, speed, and ease of use. The study shall also examine its consistency during the fasting and non-fasting states and the relevance and closeness of its findings with the conventional blood-based laboratory tests.

The technological change corresponds with the following UN Sustainable Development Goal 3: Good Health and Well-being, which prescribes the early diagnosis and optimal treatment of non-communicable diseases and the Goal 9: Industry, Innovation and Infrastructure to improve the digital health technological and optimize access to the modern healthcare facilities.

1.2 Objectives of the Study

The main objective of this study is to design and develop a non-invasive cholesterol measuring system that utilizes hand imaging data and artificial neural networks that would allow blood free cholesterol estimation in real time in order to detect early indicators of cardiovascular risk.

Specifically, this study aims to:

1. Develop a Monitoring device that would incorporate the use of camera module and controlled lighting to capture standardized image of the back of the hand to estimate the level of cholesterol.
2. Design and implement an AI model capable of processing extracted image features to classify total cholesterol levels, ensuring its performance is trained and validated against traditional blood-based cholesterol readings for clinical relevance and accuracy.
3. Develop a user-friendly mobile application, which would display the total cholesterol classification (Normal or High), and store the user information in a database.
4. Integrate hardware components such as camera, lighting system and single board computer and display to develop a small and useful prototype of real time analysis. The prototype should be suitable for use in government clinics and rural health units where access to advanced laboratory equipment is limited.
5. Test the system's accuracy, precision and usability by comparing its predicted cholesterol results against standard laboratory blood test values.

1.3 Conceptual Framework

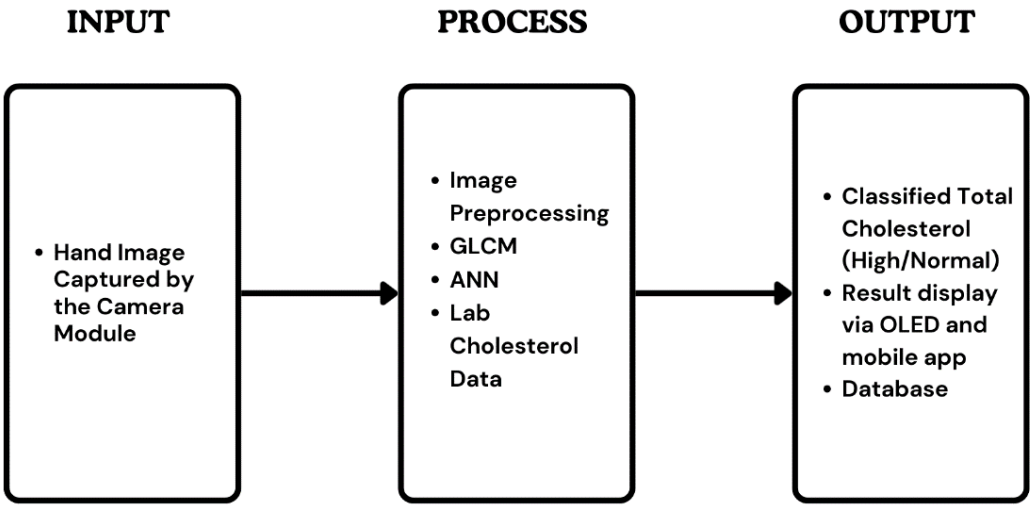


Fig. 1 Conceptual Framework of the System

Figure 1 Conceptual Framework of the System is a graphical depiction of the overall system of the proposed cholesterol monitoring system. The input phase starts with the camera unit to capture hand image, which is the primary source of data to the system. This image has given the visual contents including skin texture and lines that can be linked to cholesterol. During the process phase, the images obtained are preprocessed and this involves the normalization and removal of noise to generate images that are readable and consistent. Gray Level Co-occurrence Matrix (GLCM) is subsequently used to obtain important texture characteristics, contrast, correlation, energy and homogeneity. A validated laboratory cholesterol data are input into these features and an Artificial Neural Network (ANN) is run over them. The ANN, which is trained with reference data in the medical field, assesses the extracted features to identify patterns and come up

with correct predictions of cholesterol values. Lastly, the output stage will give the level of classified cholesterol whereby the result will be either normal or high. This classification is then presented to the user in an OLED screen and in a mobile application to provide ease and accessibility. Meanwhile, the outcomes are recorded in a database, allowing archiving records and tracking them over time to compare and analyze at a later stage.

1.4 Significance of the Study

The purpose of this study is to propose a new method of tracking the level of cholesterol based on a non-invasive technique that will enable people to better monitor their total cholesterol every day even without having to rely on blood tests conducted in labs. The study can be helpful to the following:

Senior Citizens. The study will provide the senior citizens who are highly vulnerable to cardiovascular diseases with a non-invasive, safe, and convenient way of checking their total cholesterol. The system eliminates the necessity of extracting blood painfully, which makes it easy for them to monitor their total cholesterol more frequently, which helps to prevent early diseases, preventive treatment, and improved management of health overall.

Health and Technology Sector. The study will assist the state of digital health which is increasing by demonstrating how sensor technology and Artificial intelligence could help to solve real medical problems in the current health sector that will help in the development of local technological innovation and the design of health-driven products.

Individuals of Rural Areas. The study will be applicable in the communities that have limited access to hospitals or diagnostic laboratories. With non-invasive scanning without the need of taking a complex machine, people in rural locations would be able to access the basic health information and would otherwise have to travel and spend extra money.

Future Researchers. Future researchers who embark on the study of non-invasive health monitoring systems will be able to benefit with useful information as a result of the study. It can be used as the basis of future studies on the enhancement of accuracy, other biomarkers and even to develop the system as wearable.

1.5 Scope and Limitations

This section describes scope, constraint and limitations of the proposed study. The following features will be discussed in this study:

- The study will focus on the design and development of a non-invasive prototype device that utilizes a camera module and controlled LED lighting to capture hand images for classification of total cholesterol.
- Image preprocessing techniques such as cropping, grayscale conversion, and normalization will be applied to ensure consistent data collection.
- Texture features will be extracted using the Gray Level Co-occurrence Matrix (GLCM) or Local Binary Pattern (LBP) method

focusing on contrast, correlation, energy, and homogeneity as input for the ANN model.

- There will be a basic mobile application that will display the classified total cholesterol (Normal or High) in a user-friendly interface, and securely storing data for monitoring purposes.
- The study is not intended to replace clinical diagnosis, but rather to supplement existing methods by providing a screening tool for regular self-assessment and early prevention. In settings where laboratory testing is inaccessible the device may serve as a preliminary screening method.
- The prototype will be tested with available volunteers, prioritizing senior citizens where possible, since this group has a higher prevalence of cholesterol variation.

However, this study has some limitations to be taken into consideration:

- In this study, the current version of the system will focus only on total cholesterol level Normal vs High classification. The measurement of LDL, HDL, triglycerides, and other metabolic analytes is not yet supported. Future iterations may expand to include these biomarkers as datasets and methods improve.
- The study will not support the use of any other portion of the body except the back of the hand and palm as a data collection point. The characteristics of other body parts such as wrist and earlobe

which have the ability of giving different signal characteristics are not considered in this stage of study.

- This is a system that does not diagnose or treat any condition. It is recommended that the users consult a medical practitioner about any medical matters. This must only be applied on a regular basis to be able to check by self-monitoring and creating awareness at an early stage.
- The study not consider the combined devices with wearable devices or other types of sensors like heart rate monitors or blood pressure cuffs.
- The device's results may vary depending on environmental lighting and skin tone variations, and ongoing improvements will aim to reduce these biases.

1.6 Operational Definition of Terms

- **Cholesterol –** is a waxy fatish matter present in the blood which is essential in the formation of healthy cell but excessively, causes heart disease.
- **Non-invasive –** a method of medicine or diagnosis, in which there is no penetration of the body or cutting of the skin.
- **Optical Imaging –** the process of capturing images using visible light to analyze biological tissues.

- **Artificial Neural Network –** is an artificial model which represents the human brain in terms of pattern recognition.
- **Gray Level Co-occurrence Matrix (GLCM) –** a statistical method of texture analysis that examines that spatial relationship of pixels in a grayscale image.
- **LBP –** a method of spatial domain techniques that segment an image into non-overlapping square blocks and calculate local pixel differences between a central pixel and its adjacent pixels.

Chapter 2

REVIEW OF RELATED LITERATURES AND STUDIES

This chapter reviews relevant local and foreign literature that support the development of a non-invasive cholesterol monitoring system, focusing on studies involving optical sensing technologies and artificial intelligence applications.

2.1 Foreign and Local Literature

2.1.1 Non-Invasive Monitoring

(Joseph Devakumar et al., 2019) carried out a thorough analysis of different methods and sensor systems that have already been designed to monitor the level of glucose and cholesterol non-invasively. The paper discussed optical, electrical, and biochemical methods, such as infrared spectroscopy, photoplethysmography (PPG), and biosensor-based systems, and the advantages and drawbacks of each of them in clinical and consumer applications. Particular attention was paid to the interaction of light with biological tissues, and to the effect that parameters like wavelength, depth of penetration, and variability of the skin have on the accuracy with which cholesterol may be detected. The significant issues were also mentioned in the review such as the calibration of the devices, noise in the environment, and the personal differences in physiology that restrict the popularization at the moment. This literature can be used as the basis of modern tendencies concerning the non-invasive diagnostic systems and encourage the further development of innovations in AI-enabled and real-time cholesterol sensing devices.

(Li et al., 2024) amply reviewed the newer technological developments in microneedle-based biosensors to be used in detecting analytes in dermal interstitial fluid (ISF) which is a minimally invasive method of blood sampling. The paper sheds light on the fact that microneedle platforms can be designed such that they measure set clinically measurable biomarkers, such as cholesterol, through enzymatic and electrochemical principles of detection. It elaborates on the advantages of these systems in the form of pain decrease, constant monitoring can be done, and it can be utilized with wearable electronics thus they are very appropriate when it comes to point of care diagnostics. Technical challenges like sensor calibration drift, biocompatibility, enzyme degradation and signal stability are also key issues that have put the long term reliability of the authors at bay. Their discussion validates the increasing scope of microneedle technologies in non-invasive monitoring of cholesterol levels, and the significance of compatibility between microneedle sensor with AI algorithms in future medical products in order to increase the level of interpretation and customization of the data to patients.

2.1.2 Cholesterol Level Detection

(Domínguez-Renedo et al., 2023) gave an extensive overview of the current research in the field of electrochemical sensor of cholesterol, demonstrating existing recent solutions divided into enzymatic and non-enzymatic. In the paper, some materials like nanoparticles, conductive polymers, and carbon-based nanostructures which have been employed to improve sensitivity, stability and limit of detection are discussed. It further pursues the shift between conventional benchtop instruments in a genre of wearable devices and lab-on-a-

chip ones where a high probability is upcoming in the form of point-of-care cholesterol tests. Nevertheless, the authors refer to the existing constraints as well like sample matrix interferences, biofouling, and some problems with long-term calibration. This review is especially applicable in the design of any smart biosensing device that would be effective and dependable in detecting a given level of cholesterol both in the clinical or non-clinical setting.

(Kumar & Singh, 2021)carried out a general review of optical sensing in methods used to identify various biomolecules, including cholesterol. The paper meticulously focuses on surface plasmon resonance (SPR), fluorescence, near-infrared (NIR) and Raman spectroscopy, and compares the usefulness of application in terms of sensitivity, specificity, miniaturization and having a real-time capability, in the measurement of cholesterol. It emphasizes non-invasive and reagent-free optical sensing that have proven to be an interesting replacement of conventional chemical analyses. However, other limitations to engineering as mentioned in the review include light scattering, signal calibration, and environmental variations which still require to be challenged so as to lead to better work in live conditions. These reports give a good background in the development of optical-based detector systems of cholesterol, especially when it is in non-invasive and portable form.

2.1.3 Artificial Neutral Network

In estimating the range of cholesterol and especially the LDL cholesterol, Artificial Neural Networks (ANNs) have proven to be very promising in using the routine clinical data. As (Soh et al., 2020) describe, feedforward ANN models are

more accurate than conventional formulas as they encompass nonlinear interaction between the blueprint of lipid profile compositions, which include total cholesterol, HDL, and triglycerides. Their research concluded that ANN-type predictions are more precise when it comes to high triglyceride cases and low LDL cholesterol where the traditional approach usually fails. This development brings out the applicability of ANNs in the improvement of individualized lipid profiling and real-time clinical decision support.

In the same way, (Sengupta et al., 2020) created an ANN model of assessing the total cholesterol level based on the usual demographic and biochemical data. Their multilayer perceptron network outperformed conventional regression models, and the possibility of ANN as a non-invasive, cost-effective non-invasive method of cardiovascular screening and cholesterol testing in patient groups of diverse medical histories is indicated.

2.1.4 Image Processing

The problem of efficient and accurate taxonomy of species of coffee beans, which is one of the main quality control and traceability parameters in the coffee industry, was revealed in (Klyde Adriane C. Monares, 2024). The traditional forms of identification are manual and subjective that brings about the problem of mismatch in manual assessment to the fore. The objective of the paper was to examine closely the advancements that have been made on machine learning and image processing techniques in order to identify coffee bean species. The methodology was to conduct a systematic literature review to investigate various articles that involved preprocessing techniques, feature extracting algorithms and

classification algorithms, more specially convolutional neural networks (CNNs) and support vectors machines (SVMs). The overall findings of the reviewed works indicated that the existing image processing techniques with deep learning additions are useful in improving the accuracy and efficiency of the image-to-coffee bean classification, and that some of them can achieve 95 percent accuracy rates. Despite these developments, the review identified some shortcomings such as lack of diversity of data sets, practical validation, and the need to ensure that there are standard image acquisition processes. The implication of these limitations is that more extensive research could be done in the future by building larger and more diverse datasets and developing models that are less susceptible to variations in lighting, background, and the condition of the bean.

(Geetha et al., 2023) conducted an extensive survey of different procedures of cholesterol detection that do not involve an invasive procedure, as the traditional method of blood-based testing is unpleasant and time-consuming. The paper has brought to light the issue of being dependent on the old methods that limit the regular and extensive monitoring of cholesterol levels that is most vital in the early detection and prevention of cardiovascular diseases. The paper has explored various methods such as optical sensors, biosensors and imaging-based methods and explained the principle behind each method and its practical use. Special emphasis was placed on the combination of these methods with Artificial Intelligence (AI) and Internet of Medical Things (IoMT) systems, which will allow providing round-the-clock, real-time, and patient-centered monitoring. Other challenges reviewed included calibration problems, variability of physiological

states, sensitivity and specificity constraints, and the necessity of standardized procedures in different populations. The literature underlined that although a number of non-invasive methods have demonstrated good results with respect to accuracy and feasibility, additional clinical testing and optimization of devices would be required to achieve reliable adoption. The research also offers a good starting point on how future studies can develop a more sophisticated and AI-powered non-invasive cholesterol monitoring device and promotes innovations to enhance accessibility, comfort, and real-time diagnostic performance.

(Usman Umar, 2025a), developing a non-invasive approach of measuring blood cholesterol through image processing and artificial neural networks. The study was a response to the drawbacks of traditional cholesterol tests, which is invasive, time consuming and uncomfortable to patients, limiting its ability to perform routine monitoring to detect and prevent cardiovascular disease at an early stage. The research was aimed at the analysis of the hand skin images which were taken by the smartphone cameras and using them the image process methods were used to get the features which were useful in analysis of cholesterol. The computer simulation of artificial neural network (ANN) was trained to forecast the levels of cholesterol through these characteristics and the system was incorporated into an Android-based application called Choldet to monitor the cholesterol levels in real time. The research indicated F1 of 80 percent, and the Clarke Error Grid Analysis indicated 92.98 percent of all predictions were in Zone A with high accuracy and clinical relevance. Nevertheless, the paper also emphasized the limitations, namely, the small limited data, the variability caused

by lighting conditions and differences in the color of skin tones, and the necessity of additional validation in various populations to guarantee strength. The literature is of good background to the development of AI-enabled, patient-friendly, real-time cholesterol monitoring systems, and the further research to enhance accessibility, accuracy, and generalizability of the non-invasive diagnostic tools.

The article by (Binas et al., 2023) is entitled "Jackfruit Maturity Classification Using Uniform Local Binary Pattern and Support Vector Machine), its objective being to classify the jackfruit maturity using the images of the products automatically. The authors made use of the Uniform Local Binary Pattern (ULBP) algorithm to obtain key texture information on the surface of the fruit and it was followed by a Support Vector Machine (SVM) classifier to classify the maturity stages correctly. The experiment indicated that a combination of ULBP and SVM produced high potential in detecting the various maturity of jackfruit, which evidences the efficiency of the texture concept to preprocess patterns. The studies were however restricted to agricultural purposes but did not cover such areas as medical and physiological cases, which included study of hand texture as a non-invasive cholesterol detector. This leaves an opportunity to the future research to improve LBP-based preprocessing methods in health related image analysis systems.

In an attempt to extract features of complex images better, the work by Singh (2024) attempted to construct an effective Local Binary Pattern (LBP) pattern and enhanced with deep convolutional neural networks (CNNs) to study the texture of the images. The methodology adopted a blend of traditional LBP

operators and deep feature obtained in the CNN layers, and was trained on different standard texture datasets to evaluate the system capabilities of reproducing small changes in the text. Such results depicted that this hybrid description performed better than traditional LBP and other traditional texture features in terms of accuracy in classification and immunity to changes in illumination and noise. However, the experiment failed to look at hand pictures or in medical records such as non-invasive cholesterol hence creating a gap in research with regards to applying this preprocessing technique to the analysis of hand texture in healthcare. This gap presents an opportunity to undertake a subsequent research to introduce LBP-CNN preprocessing methods into the AI-directed non-invasive cholesterol sensors.

2.1.5 Health Technology

According to (Philip et al., 2021), health technology is a term applied to any equipment, system, or software that can enhance healthcare delivery, including the diagnosis of diseases, disease monitoring, and disease management. During the last several years, more researchers have been dedicated to developing the non-invasive and intelligent technologies of health monitoring easier to implement, more convenient and available, working in real-time. Such technologies are particularly convenient to use in regions which do not have many hospitals and laboratories, or when patients need to have a close monitoring without having to go out.

(Manuel Iii & Capistrano, 2025) also created a wearable that tracks many aspects of health including the heartbeat, oxygen levels, and glucose. The

measurement of the heart rate and oxygen was precise whereas glucose was less reliable and could use some improvements. Along with this, (Philip et al., 2021) also focused on the rise in the Internet of Things (IoT)-based systems that enable a patient to monitor his or her overall health with a smartwatch and connected sensors. These devices are especially useful to people who have some chronic diseases like heart disease or diabetes. The study however raised concerns as well such as privacy of the data, complexity of the system and good internet connection. Going further in this regard, (Mbunge et al., 2021) Talked about the larger idea of Healthcare 5.0 where one uses the healthcare platforms driven by sensors, artificial intelligence (AI), IoT, 5G, and cloud computing to deliver remote and virtual care. Their work emphasized the possibility of these technologies to turn healthcare into a more patient focused, evidence based, and accessible model especially the management of chronic conditions and monitoring under distant conditions. Nevertheless, improving on these benefits, they outlined some of the impediments to implementation, which included, poor digital infrastructure, absence of policy standardizations and privacy issues plus low adoption in the underdeveloped regions. All these studies give an indication that health technology is evolving fast, and medical monitoring is becoming easy, quick, and painless particularly using non-invasive methods.

2.1.6 Hand Texture

The article by Chen, Wang, and Mao (2023) was formulated to investigate the variations in lipid metabolism on the surface of hand skin of Asiatic females in various ages. The objective of the research was to establish an age disparity in the

biochemical alteration in skin lipids and an identification of how that alteration impacts on the skin health, skin texture as well as the metabolic balance. To determine the relationship that exists between natural lipid metabolism and the broad metabolomics process, such as cholesterol regulation, the researchers employed the untargeted metabolomics technique using the high-resolution mass spectrometry data on the lipid metabolites present on the surface of the hand at different life stages. The samples were selected in the participants of the different age groups, in such a way that the biological variation was represented. These lipids were the ceramides, fatty acids, glycerides and other significant molecules that play crucial roles in skin texture and skin barrier integrity. The metabolic data were analysed with the help of the advanced statistical and pathway analysis tools which have revealed that the particular patterns of lipids were correlated with various age groups, and the abundance and type of ceramides, triglycerides and fatty acids varied significantly between the groups. These variations were explained by the changes in the secretion of sebum, moisture retention and surface smoothness- factors which directly influence skin texture. It is worth noting that the study showed that the lipid metabolism on the surface of the hand would serve as predictors of the internal metabolic variables such as cholesterol-related and general lipid regulation variables. This shows that the lipid biochemical differences that can be detected by the visual or touchable skin texture can be potentially seen, this is why the study has some weaknesses, the fact that it is not limited to biochemical profiling without including visual or touchable images of the skin. Only Asiatic females were included in the sample group, and this restricts the

generalization of the results to the rest of the population of genders and ethnicities. The research has also given a biochemical relationship between lipid composition and age but direct measure of biochemical changes to danceable alteration in hand surface texture has not been measured. The researchers could also build on this work by performing a combination of metabolomic profiling techniques with techniques of image based texture analysis and clinical measures of cholesterol in future to develop non-invasive diagnostic models of lipid-related diseases.

2.2 Foreign and Local Studies

2.2.1 Non – Invasive Monitoring

(Umar et al., 2020) developed a solution in the form of the development of a real-time non-invasive cholesterol monitoring system based on a self-made prototype and connected to a microcontroller. The system involved the capture of images of the skin especially in the hand and then computation of the same into estimations of cholesterol levels without necessarily taking bloods. The researchers also introduced wireless transmission module and used the data through embedded processing to provide readings of cholesterol. Despite these indicators of hope, the system, nonetheless, needed to be further refined in terms of optimization and the external factors that might curtail the accuracy, including the lighting and skin tone. However, it formed a concrete demonstration of concept that proved the usefulness of turning to external skin images to estimate cholesterol in real-time without the use of conventional laboratories. Equally, (Yusoff et al., 2015) invented a fingertip cholesterol meter that can estimate cholesterol levels in people with Near- Infrared (NIR) light. The mechanism of the

instrument was that it delivered NIR light at wavelengths around 1200 1949 nm and recorded the absorbance or reflectance rested on the fact that cholesterol has certain absorbance properties. The device under test was also vindicated by the researchers who compared the results of the readings done using the NIR and the readings based on standard blood cholesterol tests with positive correlation, indicating the same optical systems could be used as pain-free methods to determine the cholesterol level of the body. Nevertheless, they also admitted several weaknesses such as inconsistency caused by the thickness of the skin, the accuracy of positioning, and the device in its size too large to be used in everyday surroundings With more biochemical ideology, (Eom et al., 2020) proposed a non-invasive biosensing system based on cholesterol determination using an electrochemical saliva biosensor. The authors studied the sensor at optimized enzyme loading and platinum nano-cluster composition that made the sensor sensitive and specific to detect cholesterol in salivary samples. The sensor had a large linear response area (2-486 3M) and was selective even when there were widely used interfering agents such as glucose and uric acid. There was a good correlation with the results against the blood-based cholesterol results and thus there is high possibility of the saliva as a non-invasive method of monitoring of lipids. But to be applied in clinical setting, it was suggested to be validated in a more heterogeneous population.

2.2.2 Cholesterol Level Detection

The study by (Ni et al., 2021) proposed the use of absorption spectroscopy in detecting cholesterol in a non-invasive palm-based detection system. The

gadget applied a reagent that increased optical sensitivity to cholesterol within the skin enabling the system to determine the level of cholesterol estimates depending on how the light observed the treated palm skin surface. The authors confirmed the efficacy of the technique by comparing it with conventional gas chromatography outcomes and stated that a high correlation coefficient (r = 0.91) and an AUC of about 0.86 were observed with the technique, especially in terms of identifying patients who had preclinical atherosclerosis. These findings help sustain the notion that optical skin cholesterol sensing method has the capability of acting as a diagnostic alternative to invasive lipid tests. Nevertheless, other materials like the lighting conditions and skin elements may affect the accuracy of the system, and wider clinical testing was suggested by the authors to make the system more applicable in the real world. In another research, (Ni et al., 2022) created a superior fluorescent sensing system with a fluorescent digitonin derivative (FDD) that exclusively binds to cholesterol molecules whose number is on the skin. When this chemical biosensor is activated, fluorescence is produced or released depending on concentration of cholesterol, which makes accurate measurement possible without have to extract blood. The system was tested on 401 subjects and attained AUC of 0.9228 and 0.9422 and has high sensitivity and specificity in labeling individuals as healthy, at-risk or clinically atherosclerotic. Such a fluorescent assay method can not only be more comforting and convenient in cholesterol screening but has prospects of a real-time, point-of-care diagnosis. Although the research performed with high technical scores, its application in

portable consumer device and validation in different people were identified as future avenues to be developed.

2.2.3 Artificial Neural Network

(Mohktar et al., 2013) suggested non-invasive estimation of the level of cholesterol in the body with the help of bioimpedance analysis and artificial neural networks. The researchers gathered physiological features of the participants (body capacitance, lean mass and metabolic rate), as well as, testing various ANN training algorithms. The best one of them was the resilient backpropagation model which performed better than 82 percent. This method went ahead to indicate that the ANN models had the ability to interpret the bioelectrical data in the body and predict the cholesterol level. But then there was limitation on the population generalizability aspect of the study because only a relatively small and homogenous group was subjected to the testing. The paper has highlighted that in spite of the fact that bioimpedance is a non-invasive and potential source of input, wider validation and implementation into a mobile or wearable form factor remains elusive, (Usman Umar, 2025b)created a smartphone-based prototype in their research project, entitled, Choldet, and used hand-skin photos processed by ANN algorithms to extrapolate cholesterol levels. It employed real-time image acquisition, where ANN was trained to identify the cholesterol level and obtained a value of F1-score at 0.80 and greater than 92 percent in Zone A of ClarkeError Grid Analysis. It showed the feasibility of the utilization of simple optical imaging combined with deep learning in non-invasive tracking of cholesterol. Nevertheless, there are still some hurdles when it comes to lighting condition, differences in skin

tones, and more generalization of the models required. The authors observed that although the ANN method was effective in a controlled environment it would be improved by allowing continuous learning and calibration of the device in order to accommodate the real-world situations and the variety of users. Finally, (Argüello-Prada et al., 2025)conducted a proof-of-concept study that analyzed photoplethysmogram (PPG) signals to feed an ANN model and machine learning models to predict the level of cholesterol. They constructed their model based on PPG waveform feature extracted signals that are mostly available by fingertip pulse sensors. The study illuminated that there was a moderate relationship between cholesterol and features of PPG, which would have an effective interpretation by trained algorithms. Although creative enough, relatively small sample size and paucity in the diversity of the data presented limited the studies, which indicates a need to train and test the model on larger populations. Despite this, the research gave valuable demonstration on how optical biosignals (such as PPG) are quite signitcant in the future as a means of determining cholesterol presence in a non-invasive procedure when ANN is applied to it.

2.2.4 Image Processing

The authors studied the issue of low dermatological care accessibility in the Philippines, where most people are forced to use delayed visits or self-diagnosis that may aggravate the development of skin diseases (Casuayan De Goma & Devaraj, 2020) The purpose of the research was to develop a non-invasive diagnostic system that is able to identify typical skin diseases with the help of image processing and machine learning. To do so, the researchers

gathered the datasets of the locally common skin diseases and used the preprocessing capabilities like image resizing, relevance of noise, and enhancement of contrast to enhance the image quality. Characteristics tied to color and texture were derived out of the images and categorized with the help of algorithms as Support Vector Machines (SVM) and Convolutional Neural Networks (CNN). The findings indicate that CNN models were more effective in classifications than traditional machine learning methods and indicate the feasibility of using image processing in medical diagnostics of skin diseases in scenarios with scarce resources. However, the researchers also realized that their research had serious limitations, such as a rather limited dataset, classes of images in unequal numbers, and no experiments in real-world settings, where the variability of lighting and skin tones might lead to inaccurate results. Such lapses emphasize the need to increase image database, increase strength and create mobile-based platforms in order to make the technology more accessible to everyday use.

(Valdoria et al., 2019) created an Android-based mobile application, iDahon, which is used to detect terrestrial plant diseases by digital images processing and deep learning neural network algorithms. The research problem tackled the usual issue of Filipino farmers who tend to use manual visual inspection of produce, which can be inaccurate and time-consuming and such practices may lead to loss of yield. The primary goal was to offer an efficient and portable plant disease recognition tool that could be deployed on smartphones that could be used in the field in real-time. This methodology involved the construction of a plant leaf

image dataset, preprocessing procedures, which involved background removal and image normalization, and training a deep learning neural network model to classify plant diseases using the extracted image features. Findings indicated that the mobile app had high classification accuracy and timely feedback to users, which makes it a feasible tool to enhance the productivity of agriculture. Nevertheless, other limitations mentioned in the study were the reliance on the stable internet connectivity, limited number of species of plants covered in the dataset, and the difficulty in working with the images taken during different lighting and environmental conditions. The identified gaps indicate the necessity of the expansion of the dataset, enhancement of offline features, and additional optimization of the deep learning models to guarantee its stability in the wide field usage.

(Navarro et al., 2019) was aimed at creating an automated system to detect and classify skin diseases using digital image processing methods. The study discussed the weaknesses of conventional visual examination procedures used by dermatologists, which are subjective and may be time-consuming and attempted to offer an alternative method that is reliable, efficient, and accurate. The algorithm used in the study to extract features was called Improved Bag of Features (IBoF) algorithm, where skin lesion images are broken down into smaller patches, features are extracted with reference to each patch and the images are represented in terms of a histogram of visual words. To classify, a Convolutional Neural Network (CNN) was used to train the hierarchies of image features in space. The system was trained and tested on a dataset of images of skin lesions

and its performance was estimated based on accuracy. The findings revealed a validation accuracy of 98% and a testing accuracy of 85%, which showed the possibility of the system to help dermatologists in clinical practice. Nevertheless, the research recognized some shortcomings, including the possibility of performance variation in various types of skin diseases and image properties, the small size of the sample, and the necessity of additional studies to strengthen the study and its external validity. The existing literature forms a basis to improve automated diagnostic tools and stimulates the further evolution of AI-based software to detect skin diseases.

(Daniel et al., 2020) approached the issue of using invasive blood tests to detect the level of cholesterol only, which is uncomfortable and inconvenient to use regularly as a health check. The aim of the study was to develop a non-invasive diagnostic technique that examines the pictures of the iris as indicators of cholesterol. The research design consisted of taking quality iris images, converting them to grayscale, and subjecting them to preprocessing to increase clarity of images. Then the texture features were extracted by the help of the Gray-Level Co-Occurrence Matrix (GLCM), such as contrast, correlation, energy, and homogeneity, the input parameters of Support Vector Machine (SVM) classifier. Findings indicated that the model recorded a high accuracy rate of 96.47 percent, which implies that image-based detection of cholesterol through the use of iris is a valid method. Although the research has yielded good results, it has also reported some gaps, including the small size of the dataset, the controlled acquisition conditions, and that iris features may vary between larger populations. These

limitations demonstrate the necessity of additional research based on more heterogeneous data to verify the scalability and strength in reality.

(Umar et al., 2022)) tackled the issue of invasive and frequently expensive cholesterol monitoring procedures, necessitating lab analysis and may not be convenient as a continuous method of health management. The aim of the study was to develop a non-invasive method with the help of which the level of cholesterol can be determined based on the analysis of hands without resorting to the extraction of blood. The process took on the methodology of capturing hand-skin images and then the preprocessing of the image to make the image worthwhile. The Gray-Level Co-Occurrence Matrix (GLCM) was used to extract texture features, and it measured contrast, correlation, energy and homogeneity. An Artificial Neural Network (ANN) was then used to classify these features to predict cholesterol levels. It was found that there was a high level of correlation between the predicted and actual cholesterol values (correlation coefficient (r) of 0.95 and a high level of accuracy in Clarke Error Grid Analysis (92% ages 20-40 and 88% ages 40-60). Although these were encouraging findings, the authors of the study noted research limitations, including the small demographic sample, possible environmental noise in image acquisition, and lack of large-scale validation in non-homogeneous groups. These constraints highlight the necessity to conduct additional studies on the scalability, integration with mobile, and incorporation of more robust datasets to achieve clinical applicability.

(Usman Umar, 2025c) to solve is the ongoing invasive testing of cholesterol that involves blood withdrawal and laboratory examination, thus being

expensive and inconveniencing over an extended period. The purpose of the study was to create a smartphone-based application, named Choldet, which might be able to provide an approximate level of cholesterol in a non-invasive manner through a combination of image-processing and artificial intelligence. The research method consisted of taking images on a hand-skin using the camera on the smartphone, preprocessing the images to optimize the image quality, and deriving features pertaining to the skin texture and color difference. The latter were then inputted into an Artificial Neural Network (ANN) model which was trained and tested on a dataset of hand images with known cholesterol levels. Findings showed that the system performed well in predictive mode with high F1-scores and showed it could be a viable diagnostic instrument. But the study also recognized some of the gaps to include being limited to a small sample, reliance on controlled lighting conditions and validation not being carried out over a variety of devices and populations. These shortcomings imply that more development is required to make smartphone-based cholesterol monitoring systems robust, cross-platform, and acceptable by the clinicians.

The problem that (Mendoza et al., 2022) targeted is the problem of manual and time-consuming wood species identification in the Philippines and the fact that the classification process is very reliant on the experience and could be affected by human error. They wanted to establish a system of classifying the chosen Philippine wood species by image analysis and artificial neural networks (ANN) and create a web-based system that will make the use easy and convenient. They captured cross-sectional photos of 20 wood species using smartphone cameras

and 24x macro lenses and fixed light conditions as one of their methods. In order to obtain texture features an ANN classifier was trained and tested to automatically detect the wood species based on Local Binary Pattern (LBP) in processing the images and obtaining the features. The results showed that an average F1 score of the ensemble ANN model was 87.9, which is a high value in classification. The designed web application has been evaluated in relation to the System Usability Scale (SUS) scoring at an average of 75.6, which is regarded as the level of Good, and it was acceptable to the users and they were willing to recommend the application. The study had however gaps. The system was limited to only 20 species and all the images were made under the conditions of controlled lighting which reduced its practical use in the real world. The researchers suggested multiplication of the dataset, experiment under the conditions of variable conditions, and a mobile or offline product to increase the utility and reliability of the field.

Xie et al. (2012) carried out an elaborate research on the hand back skin texture to establish its use in personal identification and gender classification. The purpose of the study was to capture images of the high-resolution of the back hand and derive such texture-related features as Local Binary Patterns (LBP), Gabor filters and texton representations to capture the micro-patterns of the skin. These features were analyzed using machine learning classifiers including Support Vector Machines (SVM) and k-Nearest Neighbors (kNN) and compared their results in the way people are differentiated and their gender predicted. The authors came to the conclusion that non-invasive imaging with texture analysis is feasible

in biometric identification and that the texture of the hand back skin has very characteristic patterns, which can be easily classified with great accuracy. The research however did not examine the medical or biochemical meaning of hand texture as well as the potential links to cholesterol levels and neither did it put into consideration population-related differences that are relevant to the Filipino participants. This means that there is a gap in research and there is a chance that the hand texture analysis method can be applied to develop the non-invasive, AI-based cholesterol measuring appliances.

2.2.5 Health Technology

In the study of (Ignacio & Alacapa, 2025), they demonstrated that with the help of AI chest X-rays can be read to identify tuberculosis sooner and more precisely. They also discovered that the AI support readings were more stable compared to human interpreted scans and the application may reduce the price of large-scale TB testing. But they also indicated that extensive use of this system will need good access to internet, education of health workers, and acceptance by health authorities. In another study, researchers developed a mini device that would measure the blood sugar level of an individual by using near-infrared light and Raspberry Pi based without piercing through the skin. The device was effective and provided valuable results, but it continued to demonstrate validation and accuracy issues, especially across different users (Javier et al., 2020). Still, another study conducted by (Tamilselvi et al., 2020) has created IoT-based health monitoring system explicitly targeted at coma patients. In their system, they have a variety of sensors including temperature, heart rate, eye blink, SPO 2, and body

movement sensors plugged to Arduino Uno(microcontroller). These wearables monitor health data and transmit it to responsible caregivers or physicians through smartphones and on cloud. This type of monitoring in real-time enables health professionals to monitor the condition of the patient without physically being there and to take action when necessary. The system will also depend on good internet connectivity and a good cloud network which are not always easily accessible everywhere.

2.2.6 Hand Texture

In the article published by the Journal of Southwest Jiaotong University (2022), the authors aimed to develop a non-invasive procedure of cholesterol monitoring that relies on image processing. The aim of the work was to capture hand images and extract attributes associated with texture to determine changes, which could be linked to levels of cholesterol. Image texture analysis of hand texture images was done through an artificial neural network (ANN) that gave the cholesterol levels and the model was trained and evaluated on the performance measure, the root mean square error (RMSE) and correlation coefficient (r). The accuracy of the system was also determined with the help of Clarke Error Grid Analysis (EGA). Findings indicated that ANN modeling of hand texture could provide a viable, real time, non-invasive and accurate system of cholesterol monitoring. However, it was restricted only by the data on which it was founded and not by the difference between the populations, controlled light conditions, and integration of these aspects with demographic variables such as age and skin color

which could be studied in future studies and improvement of the development of AI-based cholesterol-detecting devices.

Fig. 2 Normal Cholesterol (Back of the Hand)

Figure 2 Normal Cholesterol (Back of the Hand) demonstrates the posterior side of the hand picture of a subject whose cholesterol level is normal. The skin has a smooth and homogenous skin texture with little discolor and xanthoma-like patterns. These attributes are used as initial visual attributes on image preparation and feature descriptors of the system.

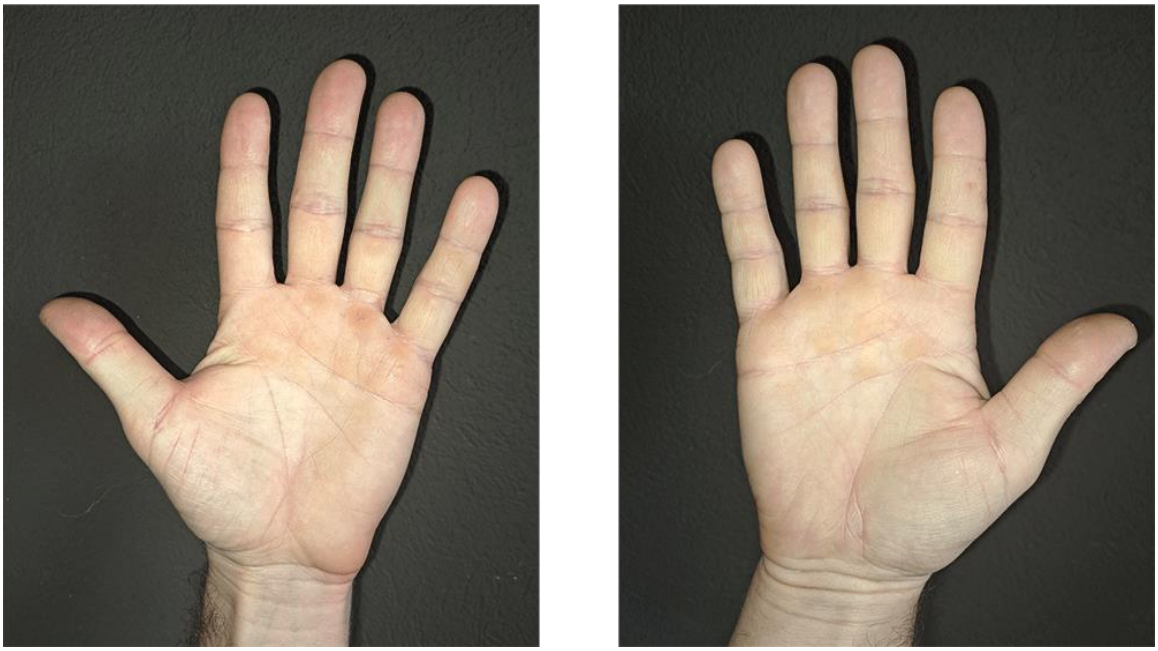


Fig. 3 Normal Cholesterol (Palm)

Figure 3 Normal Cholesterol (Palm) shows the palm of the hand of a normal cholesterol case is depicted. The palm lines and surface are characterized by uniform brightness and a consistent tone, which are under regulated lighting conditions. The ANN takes this image as a component of the training dataset in the task of normal cholesterol classification.

Fig. 4 High Cholesterol (Back of the Hand)

Figure 4 High Cholesterol (Back of the Hand) shows the reverse side of the hand of a patient with a high cholesterol level. The differences in the texture, the color shade and the lack of regularity are subtle when compared to the normal hand. The system has a texture extraction algorithm that is utilized to detect indicators of cholesterol elevation based on these visual differences.

Fig. 5 High Cholesterol (Palm)

Figure 5 High Cholesterol (Palm) shows the palm of one of the participants whose cholesterol level is high is shown. There can be uneven pigmentation or irregularities of the texture on the surface which are utilized by the ANN in recognizing the pattern during classification. Lighting control is used to make sure that the difference is not caused by external factors instead of physiological factors.

2.3 Relevance of the Study

Literature reviews and studies conducted have assisted in acquiring useful information that has assisted in determining the direction and the viability of this investigation. Previously, visualization-based detection methods have been demonstrated as useful in the medical domain. As an example, Daniel et al. proposed an iris-based image processing system, relying on the Gray Level Co-

Occurrence Matrix (GLCM) features and Support Vector Machine (SVM) to classify cholesterol levels, whereas Umar et al. developed a hand-skin imaging procedure with artificial neural networks (ANN) to estimate cholesterol level (Daniel et al., 2020; Umar et al., 2022). Equally, Umar et al. (2023) also extended this work by creating a cholesterol monitoring application on a smartphone by analyzing hand images with ANNs. These works affirmed the possibility of image processing to offer non-invasive physiological signals especially to monitor cholesterol but their limitations included reliance on controlled environments, small sets of data, and insufficient clinical validation. Furthermore, biosensing based on neural networks has also shown great performance in medical image classification and non-invasive biomarker identification (Soh et al., 2020; Sengupta et al., 2020). One of the main bases of these works is a system that is based on image capture and AI to approximate the level of cholesterol. These technologies, nonetheless, continue to be limited by weaknesses, including limited real-world testing, population diversity, and availability of deployment-ready platforms. Many of these systems are still in prototypes or lab experiments as demonstrated by both Daniel et al. (2020) and Umar et al. (2022). Just beyond these, the study of Fontanilla et al. (2019) of recognizing rice disease with image processing, Reyes et al. (2019) of the program iDahon on plant disease recognition using a mobile application, and Dela Cruz et al. (2020) on the classification of skin disease knowledge revealed the versatility of image processing in m-health and diagnostics. Nevertheless, cholesterol estimation, or artificial intelligence-driven health prediction tools did not directly preoccupy these scholars. The fact that there are no works that use these

approaches in cardiovascular monitoring indicates that more convenient and dedicated solutions in this field are required.

In this study, the researcher suggests developing a non-invasive, portable, and low-cost device of monitoring cholesterol using ANN and hand imaging. Unlike previous approaches that heavily depend on NIR spectroscopy or system-bound approaches, the proposed solution will be based on a hand imaging interface, feature extraction via GLCM, and ANN-based estimation of cholesterol, along with wireless communication to obtain real-time mobile application results. The study is done in such a way that it offers an alternative to the traditional laboratory testing and its accuracy is related to the standard cholesterol tests. The proposed system, using AI with image processing and mobile technologies, can fill the gaps outlined in the previous works, namely the unmet demand of an accessible and user-friendly cholesterol monitoring tool that can be implemented not only in the clinical setting but also in the community and remote healthcare environment.

Chapter 3

METHODOLOGY

This chapter discusses the methods and approaches the researchers used to conduct the study and reach the study objectives.

3.1 Research Design

The research design that would be most appropriate for developing a non-invasive cholesterol monitoring system is the engineering research design process. It is appropriate to employ such a design since it addresses a health-related issue in the real world and involves the progressive design, testing, and optimization of a functioning model with a combination of hardware and software.

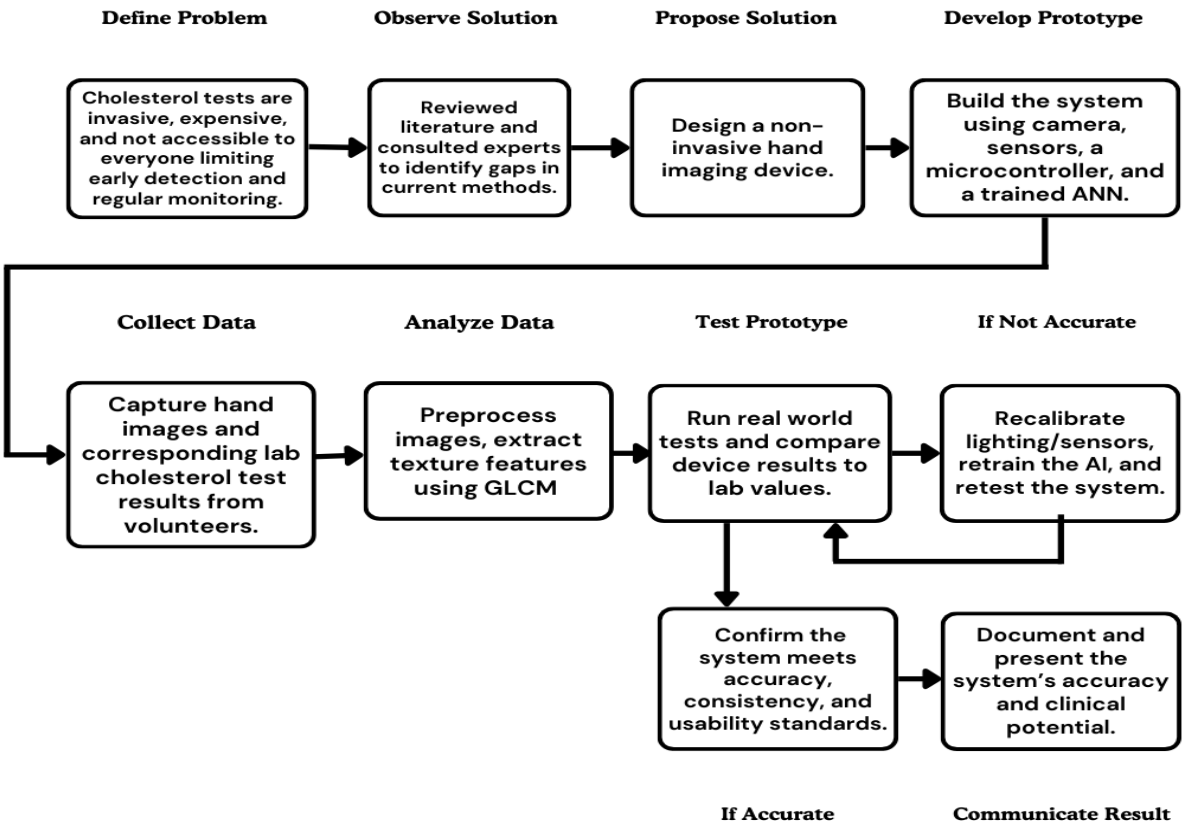


Fig. 6 The Research Process

Figure 6 The Research Process specifies every step in the process of the system development and improvements. It starts with identifying the main problem that the traditional method of cholesterol diagnostics is not only invasive but expensive, time consuming and is also unavailable to the majority of people. Having reviewed literature on the topic and using the advice of medical experts, the researchers arrive at a new solution a hand imaging device that determines the total cholesterol with the help of camera-based imaging, extraction of texture features, and an artificial neural network (ANN). A camera module and controlled LED lights are used to develop the system prototype to take pictures of the back of the hand. The images captured are preprocessed and texture attributes are obtained by the Gray Level Co-occurrence Matrix (GLCM). This prototype is tested repeatedly with volunteer subjects whose results in the non-invasive scan are compared with laboratory cholesterol result. With the results, the system is realigned and optimized to increase precision. This testing, improvement process is continued until the device yields quality and consistent results. When the necessary accuracy is reached and the potential of the suggested system will be identified as the cholesterol monitor and screening device that may be used on a regular basis through self examination and preventive medical care.

3.2 Data Collection

The data collection process will focus on generating paired datasets of hand images and laboratory cholesterol readings to train and validate the proposed non-invasive monitoring system.

3.2.1 General Plan for Data Collection

The data collection and implementation will be done in the *City Health Office under the City Government of San Juan*. At least six hundred (600) volunteer participants which will undergo two specific procedures:

1. **Hand Imaging Session –** images of the back of the hand will be captured using the prototype device equipped with a camera module and controlled LED lighting. Standardized conditions (consistent lighting, fixed hand placement, and uniform background) will be applied to minimize variability.
2. **Reference Laboratory Test –** each participant will also undergo a conventional blood-based cholesterol test (Total Cholesterol) conducted by the health office. These results will serve as the ground truth reference for training and validating the artificial neural network (ANN).

All participants will be informed of the purpose of the study, and their consent will be obtained before data collection.

3.2.2 Apparatus and Instruments

The study will employ the following instruments:

1. Prototype Imaging Device
2. Laboratory Cholesterol Test Results

3.2.3 Population and Sampling

The participants of the study will include the people taking regular cholesterol screening at the City Health Office where the device will be applied.

The elderly will be given priority because they are more prone to cholesterol changes whereas the rest of the age group will not be excluded, so as to get a more inclusive and diversified data.

3.3 Project Design Methodology

3.3.1 Block Diagram

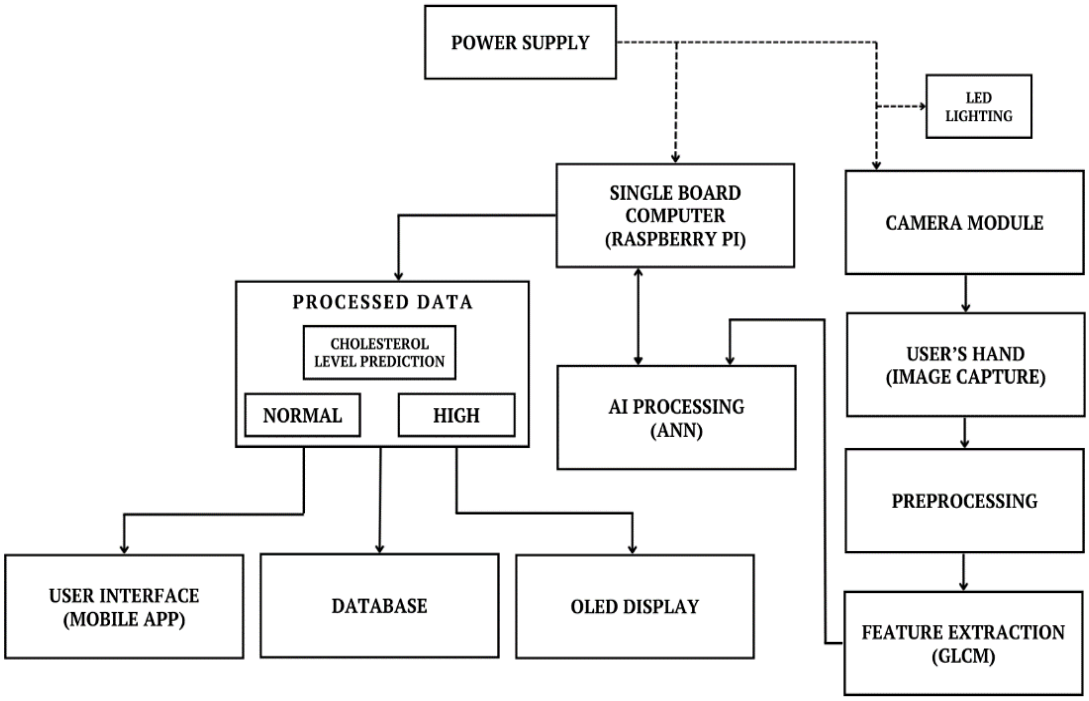


Fig. 7 Block Diagram

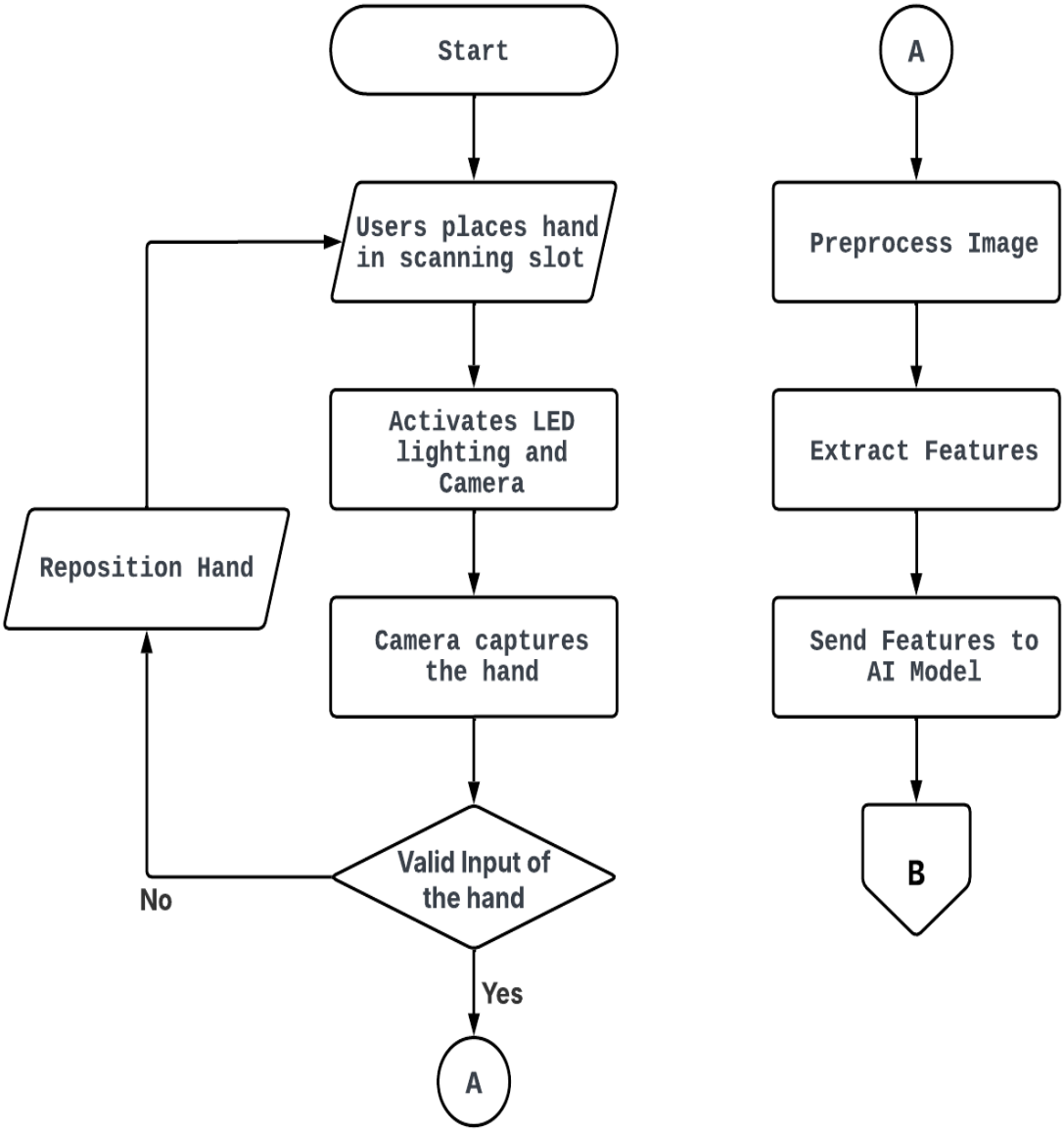
Figure 7 Block Diagram indicates the block diagram design of the proposed non-invasive cholesterol monitoring system. The system core is a single board computer (Raspberry Pi), which coordinates the hardware components including the camera module and controlled LED lighting. During operation the camera captures standardized images of the back of the user's hand under uniform illumination. The captured images undergo preprocessing (cropping, grayscale

conversion, normalization) before texture features are extracted using the Gray Level Co occurrence Matrix (GLCM) method, focusing on contrast, correlation, energy, and homogeneity.

The extracted features are processed by an Artificial Neural Network (ANN) running on the single board computer or an external computing unit, which estimates the user's total cholesterol level. These classifications are compared against laboratory results for validation.

Finally, the output stage displays the cholesterol estimate and category level whether it is Normal or High on an OLED display and transmits the results wirelessly to a mobile application, where real-time values, Normal or High classification, and historical records are accessible to the user. The mobile app also provides storage and optional cloud synchronization for long-term monitoring.

3.3.2 System Flowchart



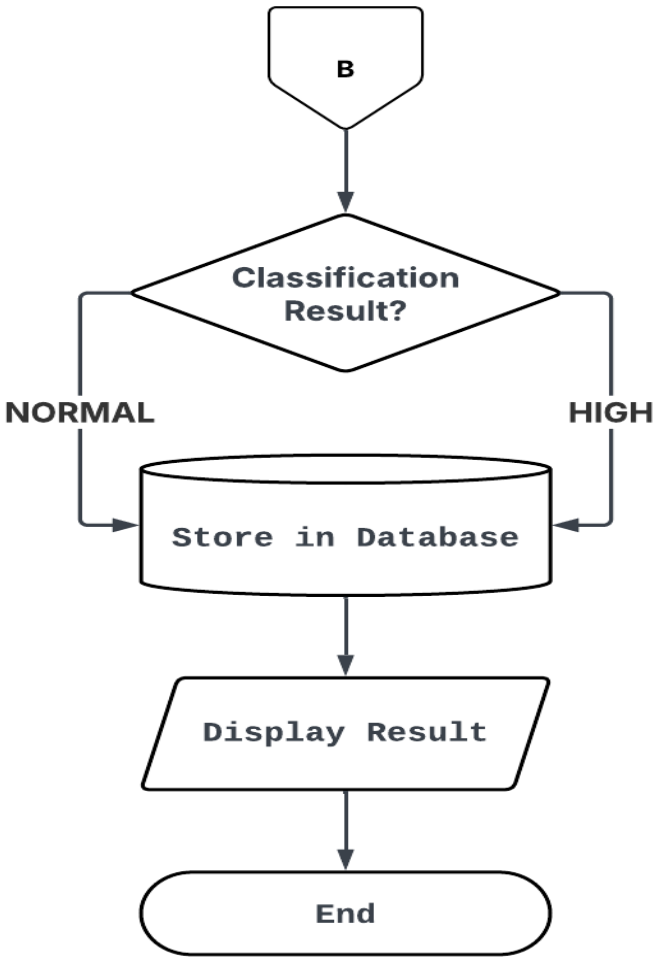


Fig. 8 System Flowchart

Figure 8 System Flowchart shows the general working process of the proposed Non-Invasive Cholesterol Monitoring System, which involves the flow of events that start with the initialization point and end with the display of results. It starts with the main components of the system being turned on and they are the single-board computer, camera module, and the controlled LED lighting that offers the system a stable illumination. The user then places the hand into the scanning slot and when they are already scanned, a picture is automatically taken with the system using constant lighting to reduce or rather eliminate variations in the

environment. The image that has been taken is subjected to preprocessing steps that include cropping, grayscale, normalization to enhance the image clarity and uniformity. The texture features are then derived with the help of the Gray Level Co-occurrence Matrix (GLCM) and perhaps, Local Binary Pattern (LBP) algorithms to derive quantifiable data regarding the skin texture and pattern. The Artificial Neural Network (ANN) is then fed with these features and the classification is done either under the Normal or the High category, on whether the cholesterol level is within the range of the Normal or the High category. The outcome is then real time shown on both the OLED screen and the mobile application and the data stored safely in the database of the system to enable tracing of the records. In general, the flowchart presents a coherent visualization of the logical process with the accentuation on the automation, accuracy, and connectivity between hardware, AI, and software units.

3.3.2.1 Mobile App Flowchart

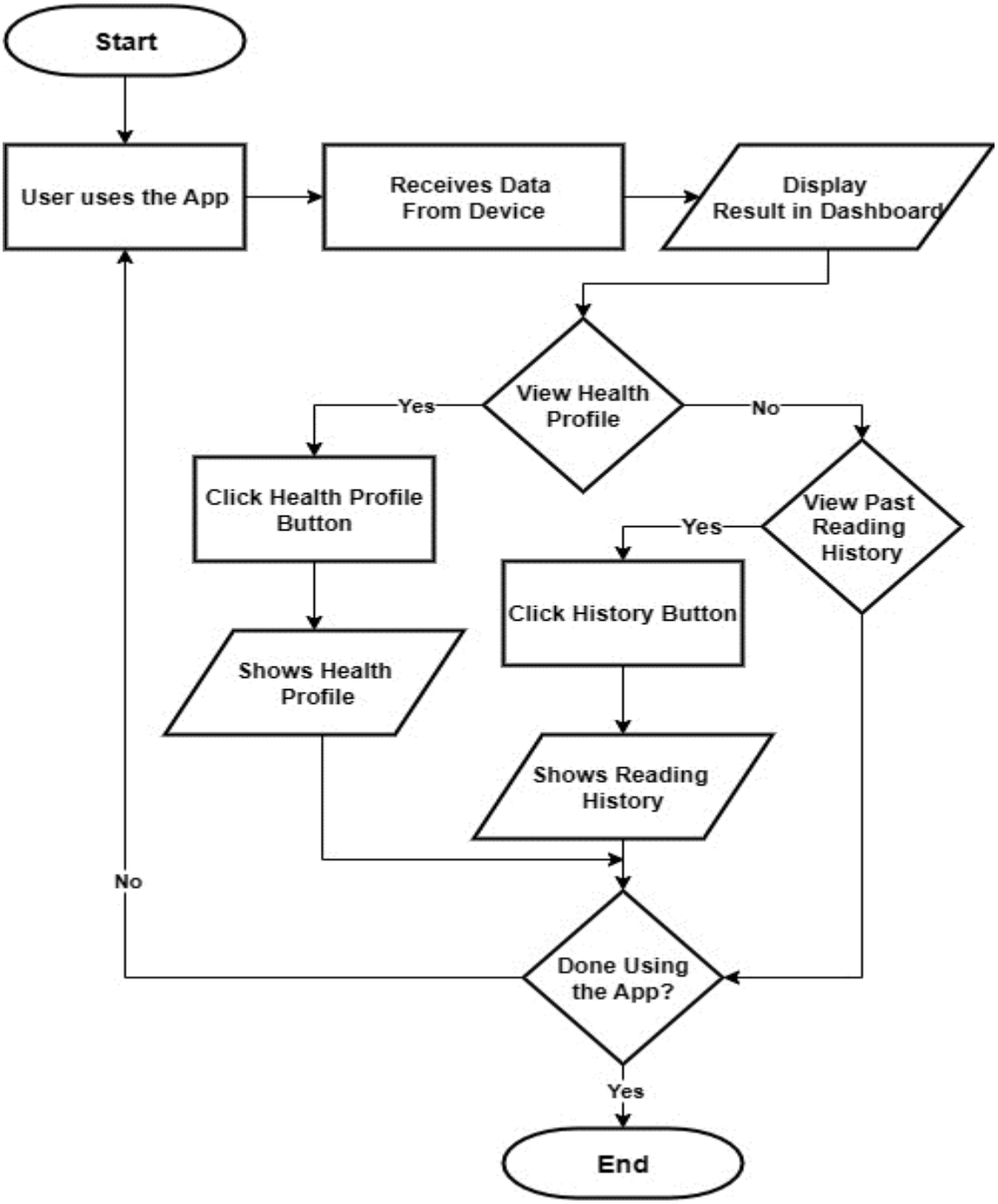


Fig. 9 User Mobile App

Figure 9 User Mobile App shows the flowchart of the workflow of the mobile application with steps which work user can follow to get access and monitor

their cholesterol results. It begins with the login and authentication process, which guarantees the security of access to health information of the user. After entering the system, the system works with the cholesterol monitoring apparatus and receives the latest results in real time. The mobile application further shows the cholesterol level of the user, either being normal or high with short definitions and recommendations on how they can keep track of their health. Other functions like historical results, profile information and tracking health trends are also available to the user to navigate through. All the records that the system contains are stamped in time and safely stored so that cholesterol variations with time can be effectively tracked and analyzed. Also, the app has a simple and friendly interface which is user-friendly to the aging population, requiring minimal technical knowledge of computer use. This flow chart illustrates the communication channel between the device and the end user which ensures convenience and reliability in health monitoring.

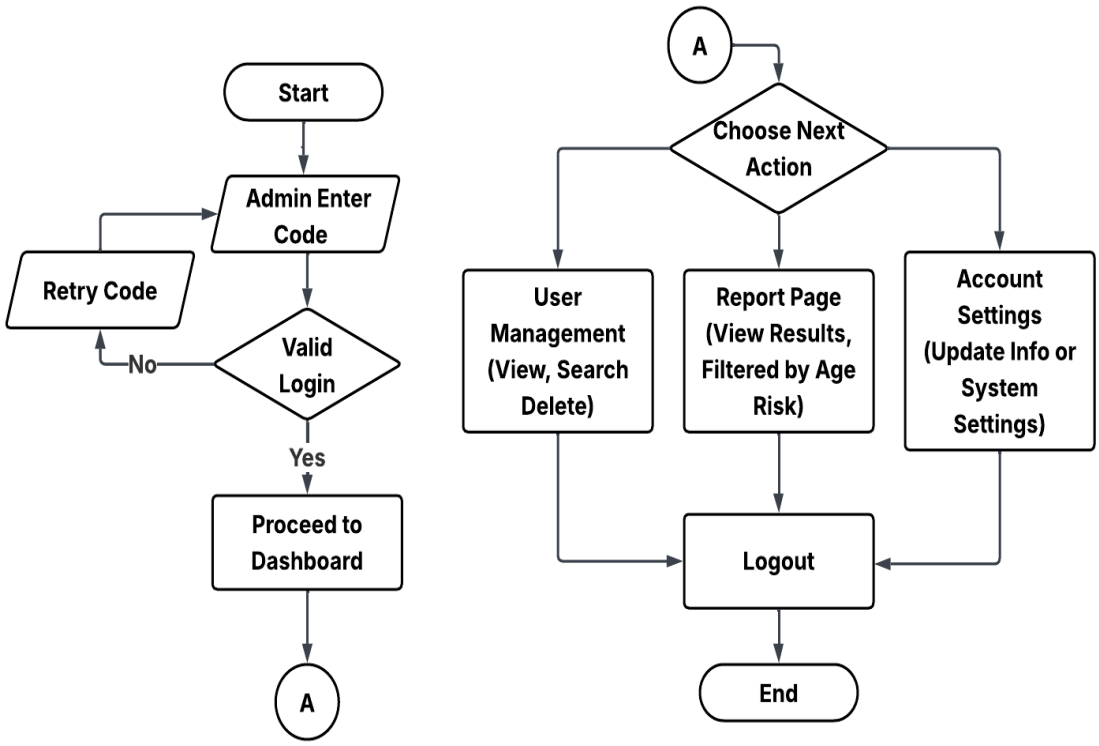


Fig. 10 Admin Mobile App Flowchart

Figure 10 Admin Mobile App Flowchart shows how it is possible to organize the management of users, monitoring of system activity, and database integrity in the cholesterol monitoring platform. This flow starts with the secure login of the administrator, as the access to the sensitive information about the system and users will only be provided to the authorized personnel. After the logging in, the admin will see the dashboard that gives the summary of overall status of the system, user activity, and cholesterol reading distributions. The administrator will be able to use the dashboard to do management functions, which include adding or removing user accounts, validating cholesterol results and monitor system performance. The admin can also create analytical reports with summarization of user data, usage pattern and accuracy outcomes of device testing. The settings of

the system configuration can also be changed to optimize the performance or refresh the AI model when necessary. The flowchart highlights the importance of the administrator in the running of the smooth functioning of the system, the security of the data, and the continuing enhancement of the monitoring system by means of the right supervision and control of the systems.

3.3.2.2 Image Capture Flowchart

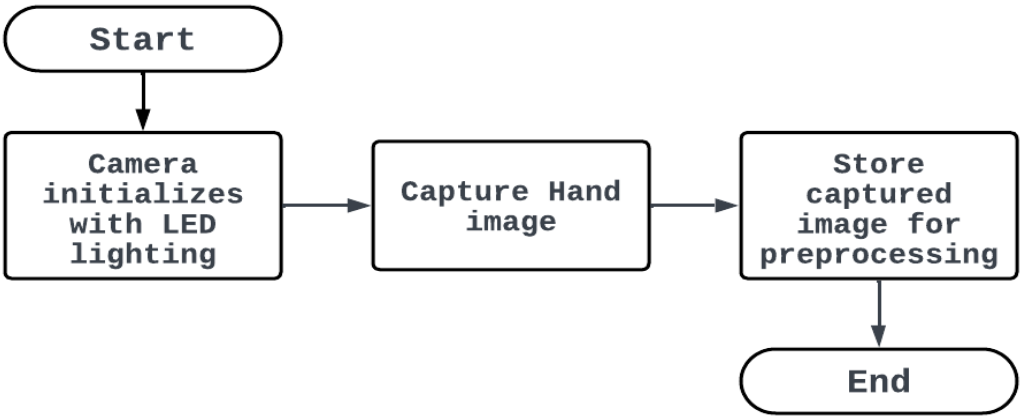


Fig. 11 Image Capture Flowchart

Figure 11 Image Capture Flowchart is the initial phase of the non-invasive monitoring cholesterol system. It begins with the initiation of the camera and the LED lighting. The LED gives adequate lighting to ensure that the picture of the hand taken is clear and stable to minimize the influence of external lighting conditions. This is done to ensure that the system is operational at the same lighting set-up each time. When the camera and LED are in place, the system proceeds to take the picture of the users hand. This hand image transforms into the raw data to be used in subsequent processes of preprocessing, feature extraction and AI based analysis. Considering that the image is the primary input,

the process of its adequate capture is also a significant portion of the entire process. After capturing the image, the image is logged in the system to undergo preprocessing. The importance of first saving the raw image is that it can be enhanced later, just by manipulating the brightness, removing noise or adding details. The image store is also useful in keeping a duplicate of the data initially stored without carrying out any processing. The flow is terminated when the captured image is successfully stored which indicates the successful completion of the acquisition stage. Now the system is now prepared to proceed to preprocessing.

3.3.2.3 AI Model Evaluation Flowchart

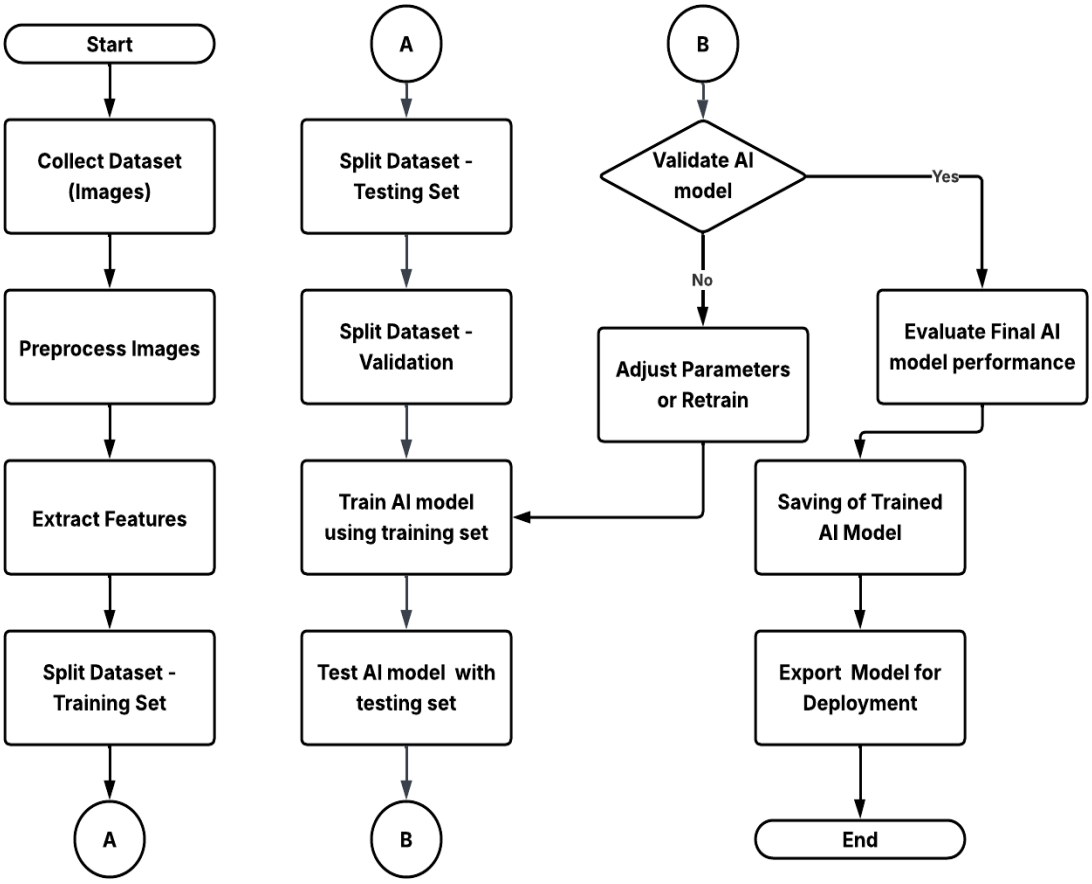


Fig. 12 AI Model Evaluation Flowchart

Figure 12 AI Model Evaluation Flowchart shows the procedural steps of the training, testing, and validation of the Artificial Neural Network (ANN) model to provide correct cholesterol classification. This begins with the collection of data in which sets of hand images and the corresponding laboratory cholesterol readings are put together in pairs. Preprocessing is done on the images to standardize the image quality by using normalization, conversion to grayscale and noise removal in order to reduce any discrepancies. The statistical texture data is analyzed with measure of texture by use of the Gray Level Co-occurrence Matrix (GLCM) which

is a useful statistical texture value, and the micro-texture is detected by use of the Local Binary Pattern (LBP) that is an important input in the ANN. The data is then split into training and testing areas, which enables the ANN to learn and subsequently test the predictive ability. The parameters of the model are constantly optimized during the training phase by reducing the error rate and maximizing accuracy, precision and F1-score statistics. Evaluation stage identifies the reliability and the strength of the model and then integrates successfully to the hardware device. This flowchart shows how scientifically rigorous and systematic the process would be in order to make the ANN model work well in non-invasive cholesterol detection in a real world scenario.

3.3.3 Requirement Analysis and Design Approach

3.3.3.1 Marketing Requirements

Table 1. Marketing Requirements

User Requirements	Specific Function	Design Approach	Standard
Non – Invasive Cholesterol Monitoring	Allow cholesterol scanning without blood extraction	Use camera and controlled LED lighting to capture hand images	ISO 13485 covers quality management systems for medical devices
Accurate cholesterol level estimation	Provide real time estimates of Total Cholesterol	Use trained artificial neural network (ANN) model for prediction	ISO 5725 covers accuracy and precision of measurement methods
Fast and easy to use interface	View results immediately after scan	Mobile App for UI and OLED display	IEC 62366-1 covers usability of medical devices
Portability and ease of access	Use device in clinics, pharmacies, or at home	Compact hardware with embedded single board computer	IEC 60601-1 covers safety of medical electrical equipment
Compatibility with multiple users	Use by adults of varying skin tones and hand sizes	Signal normalization and model training across diverse samples	ISO 24028 covers trustworthiness and reliability of AI systems

3.3.3.2 Engineering Requirements

Table 2. Engineering Requirements

Engineering Requirements	Specific Function	Design Approach	Standard
Camera Integration	Capture standardized hand images	Use camera module	ISO 12233 covers measurement of image quality
Lighting Control	Ensure consistent illumination for image capture	Use LED lighting array (White SMD LED) • 100 – 300 lumens	IEC 62471 covers photobiological safety of LED lighting
SBC control	Coordinate image capture, preprocessing, and communication	Raspberry Pi for image-heavy	IEC 62304 covers medical device software lifecycle processes
Image preprocessing	Apply cropping, grayscale conversion, and normalization	The following are the options we can use: • Grayscale • Normalization • Gaussian or Median Filter • Histogram Equalization • Region of Interest • Cropping • Edge Enhancement and Sharpening Filter • Color Space Conversion • Gabor Filtering • Data Augmentation • Wavelet Transform	ISO/IEC 25010 covers software quality requirements and evaluation
Feature Extraction	Extract important texture and color features from images	The following are the options we can use • Gray Level Co-Occurrence Matrix (GLCM): uses contrast, correlation, energy, and homogeneity • Local Binary Patterns (LBP): captures micro-textures like wrinkles and pores • Combination of LBP + GLCM + Skin Color: provides detailed	ISO/IEC 25010 covers data representation reliability

		texture and color-based analysis	
Classification AI Model	Predict total cholesterol levels from extracted numerical features	Train and evaluate various AI models to determine the best performing classifier.	ISO/IEC TR 24028 covers AI system dependability and robustness
Real time result display	View estimated total cholesterol instantly	OLED display or mobile interface connected via wireless	IEEE 802.11 covers Wi-Fi communication standards
Validation with clinical standards	Compare results against lab-tested cholesterol values	Dual testing: device and standard lab test per user	ISO 14155 covers clinical investigation of medical devices
Prototype housing and usability	Ergonomic, user-friendly hardware layout	3D-printed or molded enclosure with light isolation for sensor accuracy	ISO 10993 covers biocompatibility of medical device materials
Power Supply	Provide stable power to all electronic components	Use regulated 5V–3.3V battery, USB power, or rechargeable lithium-ion source	IEC 62133 covers safety of lithium-ion batteries or IEC 62368-1 covers safety of IT and electronic equipment powered by mains or USB Adapters

3.3.3.3 Initial Design

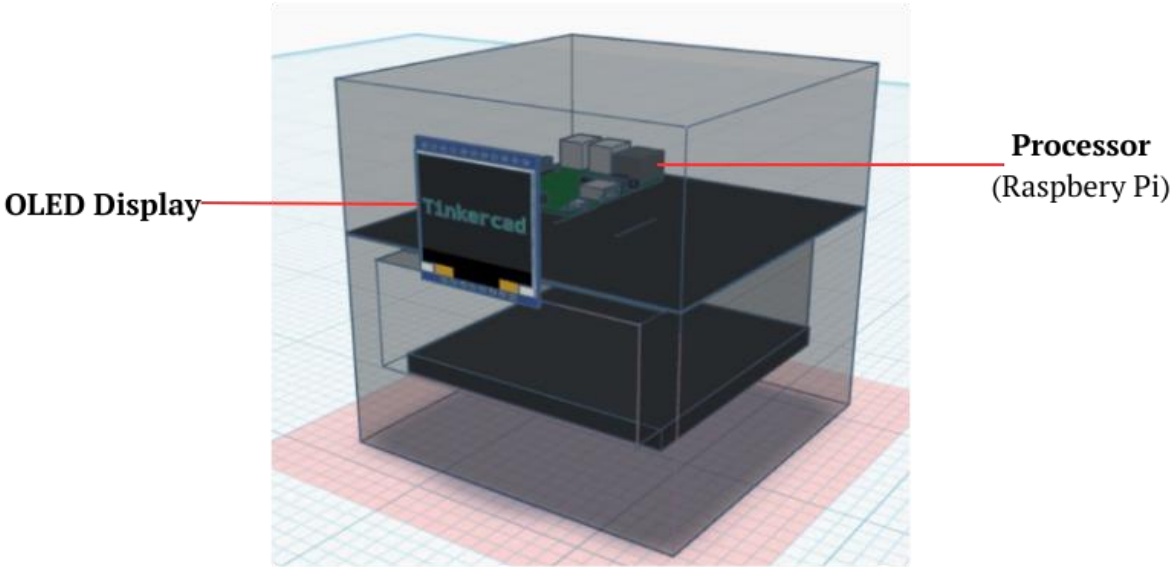


Fig. 13 Hardware Component Placing

Figure 13 Hardware Component Placing is the interior layout of the proposed non-invasive cholesterol monitoring system. The enclosure is made in such a way that it is layered enabling the suitable arrangement of components. The top part contains Raspberry Pi single-board computer that is the main controller of the system and controls image capturing, processing, and data transmission. The OLED screen is attached to the front panel and it offers a graphical interface through which real-time cholesterol levels and device status can be seen. The power supply and wiring assembly are located in the lower compartment, where they are stable and safe due to the design of the entire assembly. Such design facilitates proper space organization, elements security and convenience of maintenance without losing portability and size.

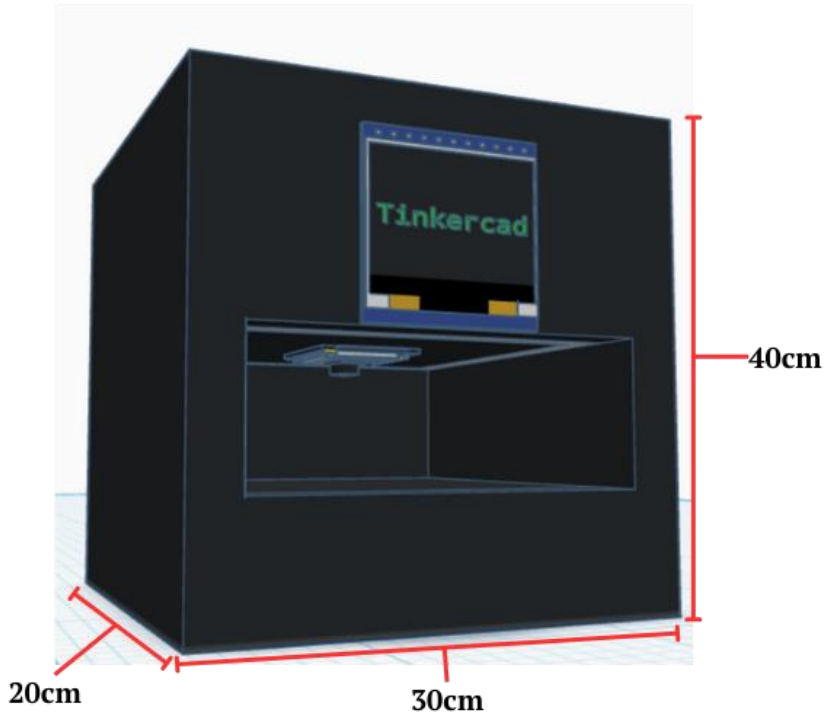


Fig. 14 Hardware Initial Size

Figure 14 Hardware Initial Size represents the prototype size and physical layout of the non-invasive cholesterol monitoring system. This symbolization shows the maximum size to be taken by the device so that it can be compact, portable, and easy to carry. The enclosure is constructed in a box form which would accommodate all the main components of the design such as the camera module, controlled LED lighting and the OLED display. The front panel has a slot to hold the hand placement where the image is always captured and an OLED screen is positioned over the front panel to show real-time cholesterol levels. The size was calculated in order to make the internal component space to be balanced with ease of transport and installation in clinics or community health systems.

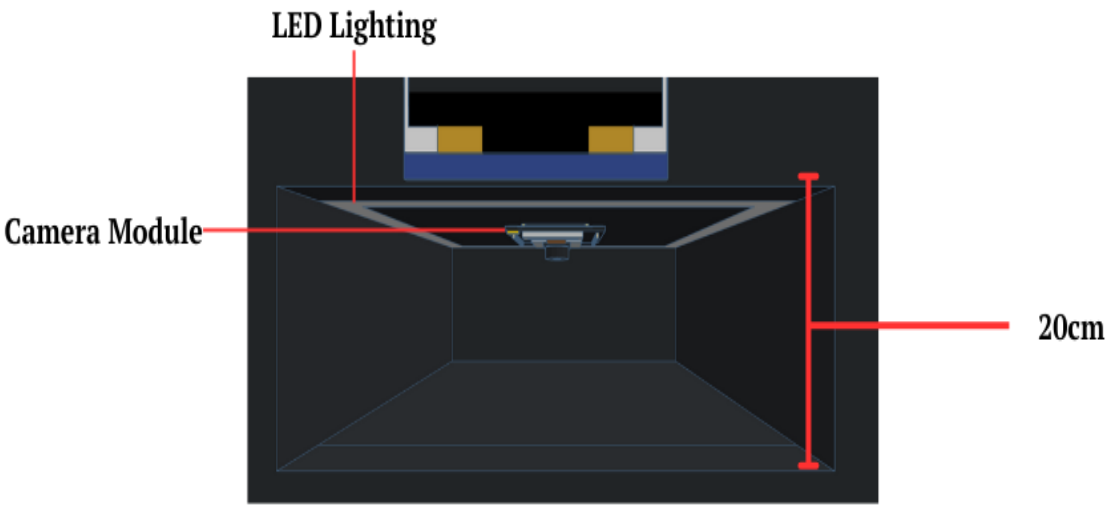


Fig. 15 Hardware Scanning Slot

Figure 15 Hardware Scanning Slot is the scanning slot pattern of proposed non-invasive cholesterol monitoring system. This is the part that the user puts their hand in the time of capturing an image. The camera module and controlled LED lighting are placed strategically in the slot to give the image of uniform lighting and reduce the impact of external light interference. The encased design ensures the consistency of the image and ensures the constant output of the light and identical positioning of the hands in each scan. The slot size was created in way that it fits the hand of an adult and it is not too big to carry around. This element is essential to obtaining standard image acquisition and having reliable data to extract features and process them through artificial intelligence.

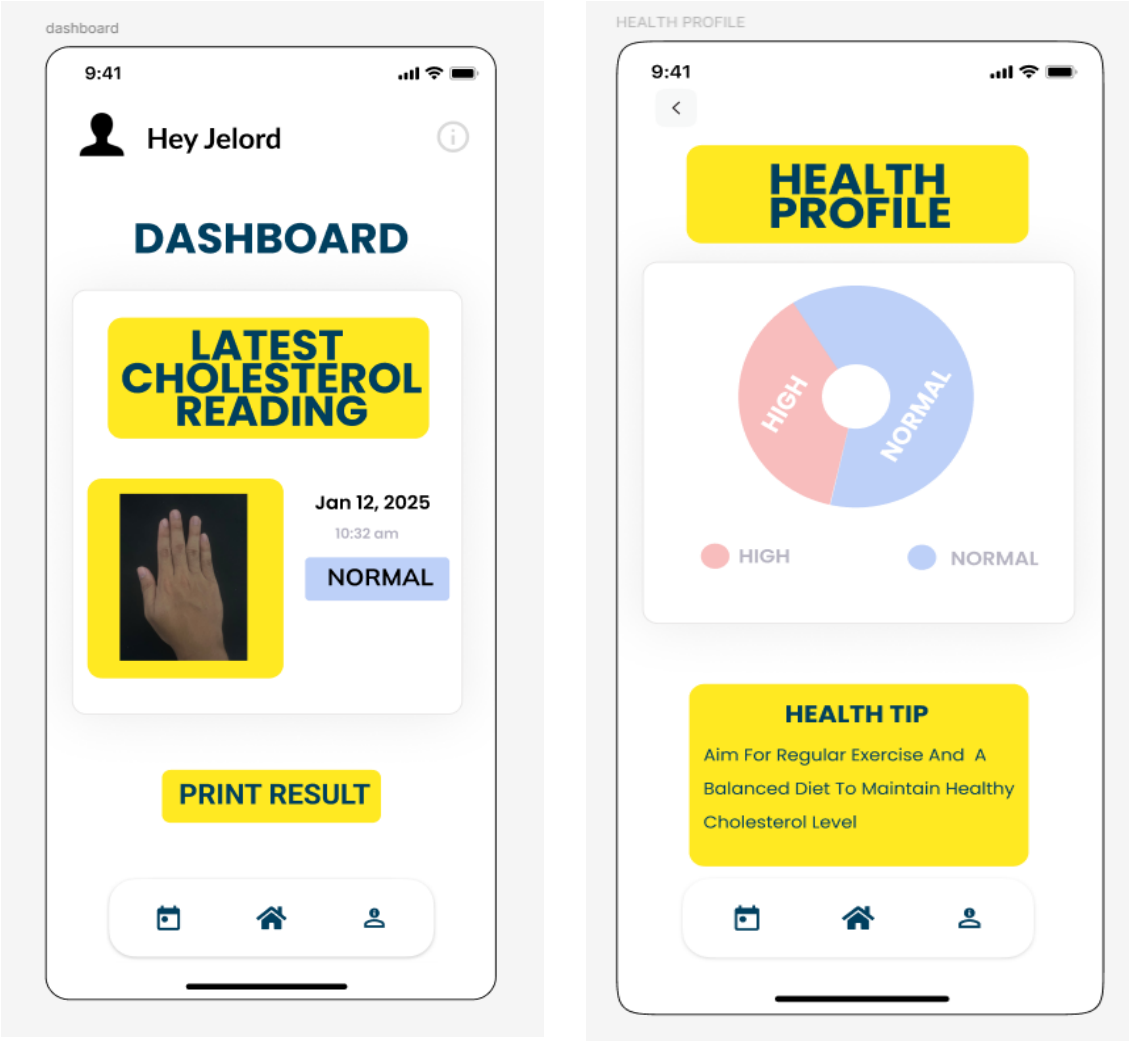


Fig. 16 User Dashboard and Health Profile

Figure 16 User Dashboard and Health Profile is the primary interface of the user-side mobile application. It shows the user his current cholesterol level, status type (Normal or High) among other simplistic health information. The dashboard is easy to navigate through to see past results along with the recent data on the device being easily visible so that the person can monitor himself or herself.

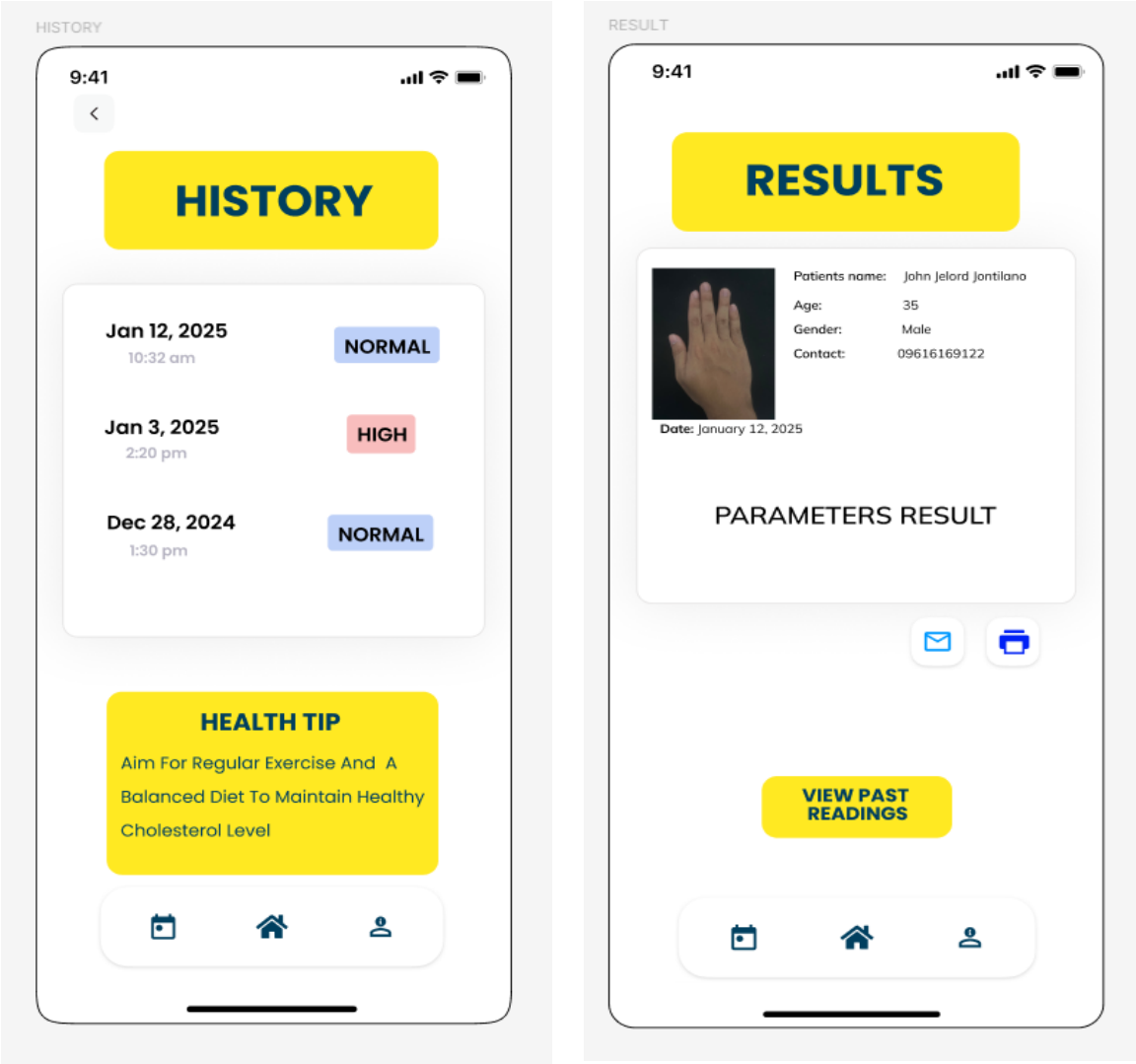


Fig. 17 User History and Reading Result

Figure 17 User History and Reading Result represents the part of the mobile application which contains the past cholesterol readings of the user. Every record contains the date and time and the type of cholesterol level. This option enables the users to follow their patterns of cholesterol with time and see their progress in care of their health. Additionally, the interface has a print or export a result option that lets the user create either a hardcopy or electronic copy of his or her cholesterol records to record himself/herself for medical consultation.

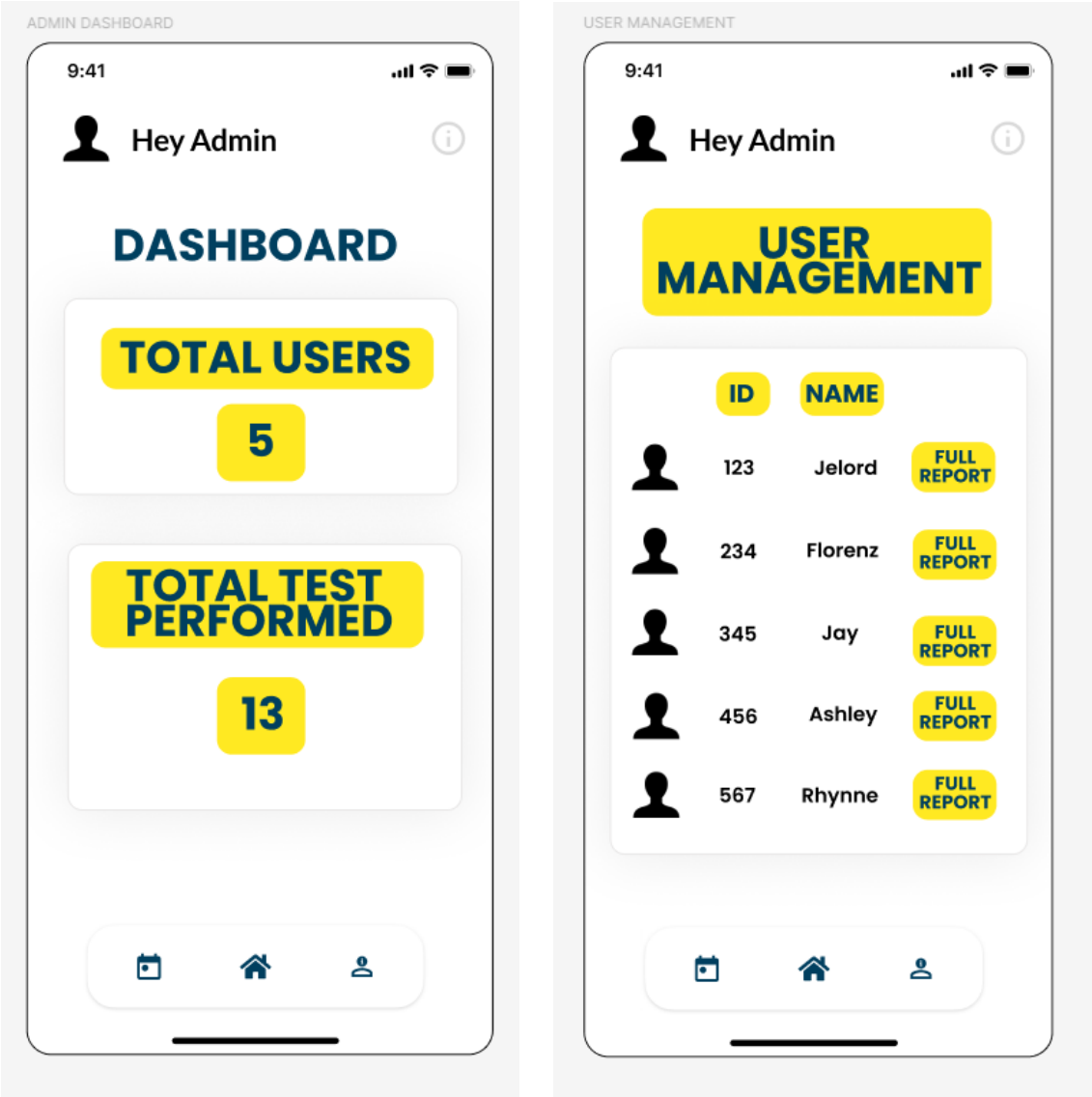


Fig. 18 Admin Dashboard and User Management

Figure 18 Admin Dashboard and Admin User Management is the administrative interface of the system. It allows registered users to be tracked by authorized individuals in monitoring them, make and manage their accounts, and access their reported cholesterol readings. The dashboard gives a summary of the activity in the system and is used as a control panel to ensure that the data records are accurate and well-organized.

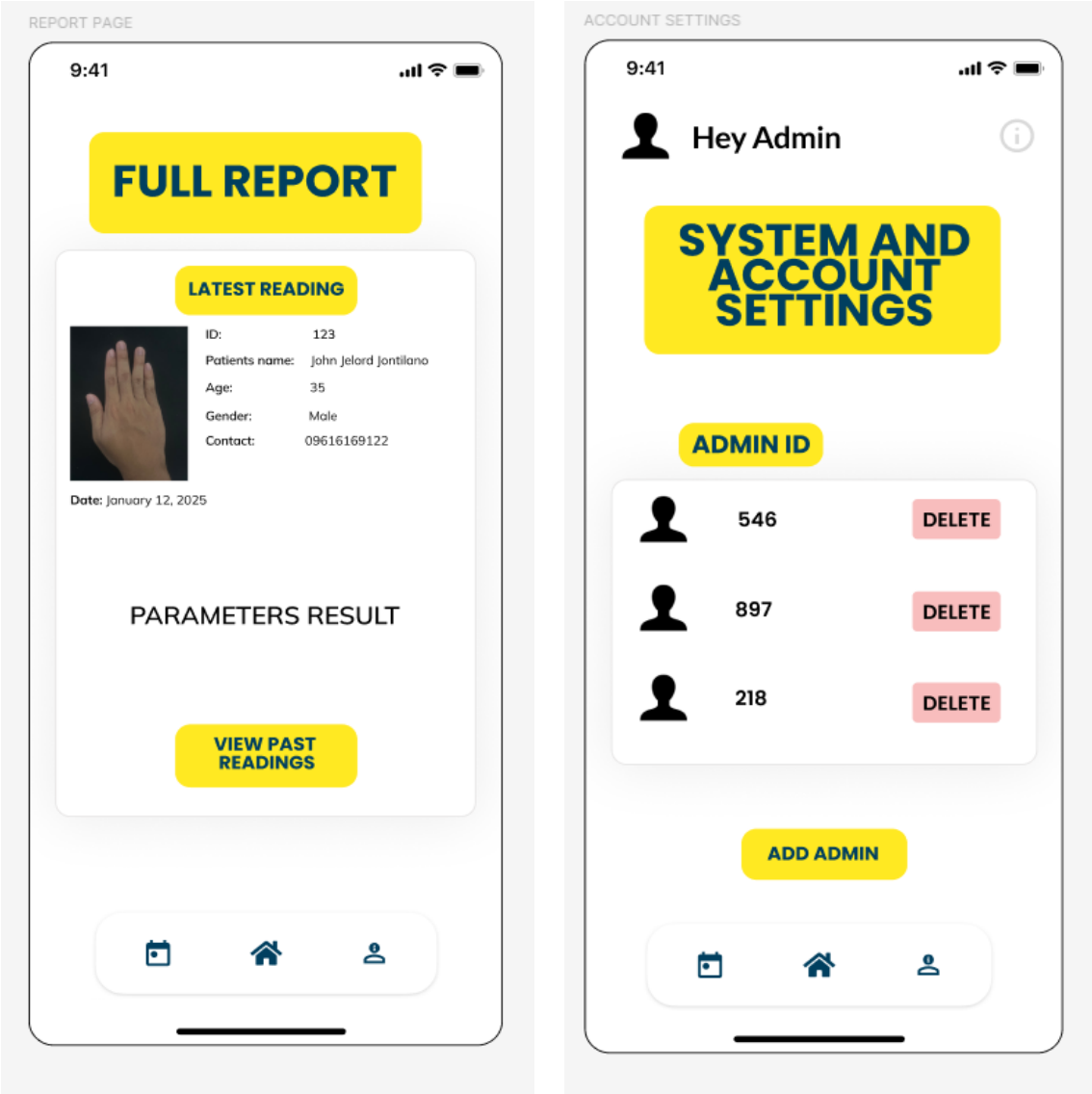


Fig. 19 Admin Report Page and Account Settings

Figure 19 Admin Report Page and Account Settings provides the access of the administrator to report generation and account settings. With the help of this page, the admin will be able to make summary reports, review usage statistics, and modify account credentials or system parameters to make sure that the cholesterol monitoring platform runs safely and effectively.

3.3.3.4 User Procedure

1. Power On the Device

- Switch the device on and leave the LED lighting system to stabilize to produce a stable lighting to ensure that clear images of the hands can be made.

2. Connectivity to Mobile Application

- Ensure that you go wireless to the screening device using the dedicated mobile application.

3. Place Hand in Scanning Slot

- To take a picture, place a hand on the scanning slot of the device with a flat and consistent position of a hand.

4. Initiate Scan

- The scan of the hand image can be initiated by pressing the scan button on the device or the mobile application.

5. Image Capture and Preprocessing

- The device captures the image of the hand under constant the camera and LED light conditions. The image is afterwards real-time pre-processed.

6. Artificial Intelligence Analysis and Feature Extraction

- The textural or the features attributes are received and the AI system (ANN) is to provide the output of these attributes into the classes of either a cholesterol level of a Normal one or a cholesterol level of a High one.

7. View Results on Mobile App

- The output of classifications is presented on the mobile application screen with the previous scan record in case it exists.

8. Data Logging and Export

- The scan is automatically added to the application history that is used in the future. The users can share and export the reports as per the need.

9. Power Off the Device

- After scanning the device, it is recommended to close it correctly so that it would save the energy and reach the highest hardware life.

3.3.4 Software Design Methodology

The study will use the Agile Software Development Model in creating the system. Agile allows flexible and step by step development, where each part of the project such as image data collection, AI model training, and mobile app integration is done in short cycles. After every cycle, the output is tested, and any problems found are fixed after the testing. This approach helped the team improve the system continuously based on testing and feedback. It made the development faster, more organized, and ensured that the final system works accurately and is easy for users to use.

3.3.4.1 Classification Model Algorithm

Its primary classification model is an Artificial Neural Network (ANN), which is used to compare the hand image of the user and decide whether he has a Normal or High cholesterol level. The ANN works based on the visual features,

including texture, tone and brightness, obtained out of the image and processed through a series of layers, i.e. input, hidden and output layers. Every image is first subjected to preprocessing before analysis, which consists of grayscale conversion, normalization, and resizing to ensure the same quality of data. The model is optimized to attain a consistent accuracy and it is tested to provide consistent performance in predicting the level of cholesterol.

The software development is implemented in Python as the primary programming language, and the ANN is constructed and trained with the help of TensorFlow and Keras, and the image is processed with the help of OpenCV. The model is trained on Raspberry Pi, which operates the camera, image data processing, and results display. A Flutter based mobile app is built to log, store and visualise real time cholesterol through a user friendly interface with the help of Firebase.

The general algorithm consists of three key steps, they are preprocessing, during which the image is processed and is ready; model processing, during which the ANN makes the classification; and testing, during which the predicted outcomes are verified against the standard cholesterol outcomes. This software and intelligent-based algorithm integration has created a reliable, non-invasive, portable, and cholesterol monitoring system.

3.3.4.2 Software Tools and Platforms

In order to create the non-invasive cholesterol screening system, the use of various software and platforms was made to combine the elements of artificial intelligence, image processing, and mobile functionality into a single

system. The major program language (PL) is Python, which is selected because of its flexibility and vast libraries which facilitate deep learning and computer vision. An Artificial Neural Network (ANN) model was created, trained, and tested with the help of such important libraries as TensorFlow and Keras that perform computational tasks of training and prediction. The preprocessing of the hand images was done using openCV and involved cropping, resizing, converting to grayscale and normalization. It was the tools that allowed successfully training the ANN model and deriving pertinent image features that would be necessary to properly classify cholesterol.

ANN model is trained and deployed on a Raspberry Pi 4 which is the central processing unit of the system. The Raspberry Pi controls the hardware parts like the camera module and LED light and runs the AI model to real-time detect cholesterol. Portable health applications are perfect applications of the embedded platform due to its small size, low price, and performance. In order to get better accessibility, a mobile application was created with Flutter which is a cross platform development framework that facilitates support for both Android and iOS platforms.

The app is linked to Firebase database, which offers a safe and a synchronized cloud-based data repository between the Raspberry Pi and the smartphone of the user. This will help users check their cholesterol levels easily, their health trends and view their past records in real time. In sum, all these software and sites will make the system work efficiently, providing quick output image processing, consistent cholesterol classifier, and easy interface to monitor cholesterol without the use of invasive methods.

3.3.4.3 System Algorithm and Processes

The system takes a three-step algorithm process which involves preprocessing, model training, and testing and validation. The classification of the cholesterol level is accurate, efficient, and consistent in each of the stages.

During the preprocessing stage, the image of the captured hands is subjected to various imaging enhancement algorithms in order to get the hands ready to be analyzed. To enhance the computation complexity, the images are converted to grayscale and normalized to make the brightness and contrast constant. Noise in the images is removed using Gaussian or median filters and all images are scaled to the correct size needed by the Artificial Neural Network (ANN). Other procedures, like histogram equalization, ROI cropping, edge enhancement, and Gabor filter promote the texture of skin features, and Wavelet Transform (DWT) and data augmentation enhances the texture and model strength.

The system employs Local Binary Patterns (LBP) and Gray Level Co-occurrence Matrix (GLCM) in extracting features.

- LBP is able to capture fine textures including wrinkles and pores.
- GLCM examines the relations more generally, and calculates such features as contrast, correlation, and energy.

When used together with the analysis of skin color, these techniques can give detailed data on both micro and macro skin textures which can be the evidence of cholesterol accumulation.

The ANN is used to analyze the preprocessed and feature enriched images in the model processing phase and the ANN is made up of the input layers, hidden layers, and the output layers. The ANN is trained to identify cholesterol levels as normal or high using the patterns of texture and color using the backpropagation technique. The system is also flexible, and depending on which approach yields the best accuracy and performance in the evaluation process, the model or algorithm can be changed or substituted. Lastly, in the testing and validation phase, the trained model will be tested on unseen data and the parameters of accuracy, precision, recall, and F1-score will be used to evaluate performance. The model is tested and deployed on Raspberry Pi to monitor it in real-time. The device does the processing on-site and sends data with cholesterol to a Firebase to a mobile app, indicating and storing the data.

This algorithmic procedure, in general, involving advanced preprocessing, LBP and GLCM feature extraction, and adaptive ANN modeling, guarantees accurate, reliable and flexible non-invasive cholesterol detection system.

3.4 Testing Plan and Procedures

Table 3. Hardware Table

Test ID	Test Objective	Input/Condition	Expected Result	Actual Result	Pass/Fail
HW-1	Verify camera image clarity	Capture hand under LED lighting	Hand region clear, no shadows		
HW-2	Check LCD display	Send sample text via single board computer	Clear readable text shown		
HW-3	Verify wireless module	Transmit data to mobile app	Data received within ≤5 sec		

HW-4	Verify LED lighting intensity	Supply 3.3V to LED	Bright illumination within set lumen range		
HW-5	Check power stability during operation.	Run device on 5V–3.3V supply for continuous 30 minutes.	No flickering, restarts, or power drops occur.		

Table 3 Hardware Table is a methodical design that will prove the vital physical appliances of the device, including the camera, the light of LED, wireless module, and power supply. It also checks that these hardware components can perform as per the expected conditions to give clear, stable and reliable input data to be analyzed by the downstream.

Table 4. Software Table

Test ID	Test Objective	Input/Condition	Expected Result	Actual Result	Pass/Fail
SW-1	Test image preprocessing	Input raw hand image	Cropped, normalized, enhanced image		
SW-2	Validate feature extraction	Input standardized image	Correct GLCM/LBP metrics extracted		
SW-3	Mobile app history logging	Run 5 scans in sequence	All scans saved with timestamp		
SW-4	Validate ANN model prediction	Input preprocessed features	Predicted cholesterol value and classified level		

Table 4 Software Table consists of a detailed checklist used to confirm the presence of software processing pipeline, which includes raw image preprocessing, feature extraction, data logging inside the mobile application, and prediction results inside the AI model. These tests ensure that the software components properly decode and process input data that is required to execute the overall functionality of the system.

Table 5. Integrated (Operational) Table

Test ID	Test Objective	Input/Condition	Expected Result	Actual Result	Pass/Fail
INT-1	Hand Scan Process	User places hand then scan starts	System captures and processes within 10 sec		
INT-2	ANN Classification Flow	Processed Hand image	Predicted Cholesterol Category		
INT-3	Print/Export function	User taps “Print Result” button	File exported/shared correctly		
INT-4	Database reliability	perform 20 scans	All data saved without loss with timestamp		

Table 5 Integrated (Operational) Testing Table confirms the end to end user experience of hand placement to cholesterol classification and data management. It balances perfectly with hardware and software, high processing speed and proper exportation of data and hence makes the system viable and easy to use.

Table 6. Integrated (Performance) Table

Test ID	Test Objective	Input/Condition	Expected Result	Actual Result	Pass/Fail
PF-1	Classification accuracy	Classified Values and Lab Results	≥90% match with lab results		
PF-2	Response time	Full workflow from scan to output	≤ 5 sec delays		
PF-3	Consistency	Repeat scan 5x under same condition	Stable results (≤ 5% variance)		

Table 6 Integrated (Performance) Table sets quantitative standards that measure the major performance indicators like the classification accuracy (to reach 90 percent agreement with lab results) response time (not more than 5 seconds) and repeatability of measurements. This will ensure the accuracy of the testing of the device and its efficiency in operations and therefore one can be assured of its clinical relevance.

Table 7. Usability and Reliability Testing Table

Test ID	Test Objective	Input/Condition	Expected Result	Actual Result	Pass/Fail
URT-1	Stress Testing	20 Consecutive Scans	No crashes, ≤ 2% error drift		
URT-2	Low light environment test	Scan hand in darker setting	Still detects hand reliably		
URT-3	App navigation usability	10 users navigate app	≥ 90% report easy use		
URT-4	Setup simplicity	User sets up device + app	≤ 5 min average setup		
URT-5	Check consistency across states (Fasted vs Nonfasted)	Scan same user fasted & nonfasted	Same output (Normal or High) in both states with only minor variation		

Table 7 Usability and Reliability Testing Table concentrates on the real-life scenarios such as lengthy stress testing, working in low-light conditions, user-friendliness of the app, installation ease, and surgical stability of measurements in varying states of physiological conditions. Such tests guarantee that the gadget remains strong and acceptable in real-life situations other than in the laboratory environment.

Table 8. Algorithm Comparative Table

Test ID	Algorithm	Test Objective	Evaluation Metrics	Expected Result	Actual Result
APT-1	Preprocessing: Cropping + Grayscale + Normalization	Establish baseline preprocessing performance	Accuracy, Precision, Recall, F1-score, Processing time	Acceptable detection with moderate accuracy and fast processing	
APT-2	Preprocessing: + Noise Removal (Gaussian Filter)	Assess improvement from noise reduction	Accuracy, Precision, Recall, F1-score, Processing time	Higher accuracy and precision compared to baseline	
APT-3	Feature Extraction: GLCM only	Measure performance using texture features alone	Accuracy, Precision, Recall, F1-score,	Moderate performance but limited	

			Processing time	feature discrimination	
APT-4	Feature Extraction: GLCM + LBP	Combine texture + local patterns for improvement	Accuracy, Precision, Recall, F1-score, Processing time	Significant improvement in recall and F1-score	
APT-5	GLCM + LBP + Skin Color	Evaluate multi-feature approach	Accuracy, Precision, Recall, F1-score, Processing time	Best performance among feature extraction methods	
APT-6	AI Model 1 (Simple ANN)	Test classification on shallow network	Accuracy, Precision, Recall, F1-score, Processing time	Good accuracy but susceptible to misclassification on noise	
APT-7	AI Model 2 (Deep ANN with Dropout)	Evaluate advanced deep learning model	Accuracy, Precision, Recall, F1-score, Processing time	Highest performance, more robust against variations	

Table 8. Algorithm Comparative Table is a comparative table with the analysis of the system utilizing various preprocessing, feature extraction, and ANN setups to analyze the performance of the system. It points out that combinations of GLCM, LBP, and color features of the skin, and Deep ANN model enhance accuracy and robustness. Such comparative outcomes would provide the final model setup with dependable and optimized cholesterol level classification performance.

Table 9. Comparison of AI Model Performance Metrics with Previous Study

Metrics	Expected Result	Previous Study Result (Umar et al .(2025))	Our Study (Actual Result)	Remarks
Accuracy	3 to 5% higher than previous study	82.14%		
Precision	3 to 5% higher than previous study	71.43%		

Recall	3 to 5% higher than previous study	90.91%		
F1 Score	3 to 5% higher than previous study	80.00%		

Table 9. Comparison of AI Model Performance Metrics with Previous Study is a comparative table that lists the important classification metrics and in this way compares the proposed AI-based cholesterol classification model with prior research conducted by Umar et al. (2025).The performance of the developed AI model will be evaluated using essential classification metrics derived from the confusion matrix: accuracy, precision, recall (sensitivity), and F1 score. These metrics are formally defined as:

- **Accuracy** Ratio of correct prediction to the total number of cases.
  - **Formula:** Accuracy = $\frac{TP+TN}{TP+TN+FP+FN}$ (1)
- **Precision** Fraction of correct positive predictions out of the total number of positive predictions.
  - **Formula:** Precision = $\frac{TP}{TP+FP}$ (2)
- **Recall** Percentage of the true positives that have been detected out of all the true positives.
  - **Formula:** Recall = $\frac{TP}{TP+FN}$ (3)
- **F1-Score** balances the two metrics, it's a preferred summary for out AI
  - **Formula:** F1-Score = 2 × $\frac{Precision \times Recall}{Precision + Recall}$ (4)

3.5 Design Constraints

3.5.1 Cost

- The total cost of components should remain affordable for prototype development and potential deployment in community health centers.
- Use of commercially available, off-the-shelf parts is preferred to minimize expenses while ensuring quality.

3.5.2 Portability

- The device should be small and lightweight enough to be placed on clinic desks or carried for community health visits.
- It should function without requiring bulky external equipment, allowing convenient use at home or in rural areas.

3.5.3 Lighting Control

- Controlled LED illumination must provide uniform lighting conditions for every scan, reducing shadows and reflections.
- The design should minimize the influence of ambient light by using an enclosed or semi-enclosed scanning chamber.
- Consistent lighting ensures reliable image quality for feature extraction and ANN processing.

3.5.4 Sustainability

- The system will be powered directly via a USB adapter connected to an outlet, removing the need for disposable batteries.

- The design should support long term use with minimal maintenance and easy replacement of components.

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College of Computer Studies and Engineering
Computer Engineering / Electronics Engineering Department

THESIS PROPOSAL / DESIGN PROJECT
REVISIONS COMPLIANCE FORM

PROJECT TITLE: Non-Invasive Cholesterol Screening Device Using Hand Imaging and Artificial Neural Networks		
Program: **Bachelor of Science in Computer Engineering**	**Course/Section: CPE C403 – 401G**	**School Year:** **2025 - 2026**
Subject Adviser: Engr. Rosalina Estacio		
Proponents: **1. Derilo, Ashley Shane B.** **2. Francisco, Florenz Jay B.** **3. Jontilano, John Jelord M.** **4. Villas, Rhynne Charz M.** **Technical Adviser: Richard Ramos, PhD**		

List of Recommendations	Revision/s Made	Page or Part of Project reflected
Revision of Title	*Changed "monitoring" to "screening" and removed "senior citizens.".*	*Title Page*
Specify LED lighting type and specs	*Added detailed LED specifications under Engineering Requirements.*	*Ch. 3.3.3.2 (P.55)*
Clarification in database usage	*The database used is Firebase. It stores the following data:* *User Information, Cholesterol Test Data, and Admin Accounts.*	*Entire Paper*
User App and Admin App	*Added wireframes for both User and Admin applications.*	*Ch. 3.3.3.3 (P.60, 61, 62, 63)*
Alternative preprocessing (besides GLCM or combination)	*Included other possible preprocessing techniques we can use under Engineering Requirements.*	*Ch. 1.2 (P.4)* *Ch. 3.3.3.2 (P.55)*
Consider alternative preprocessing technique	*Included alternative preprocessing techniques under Engineering Requirements.*	*Ch. 1.2 (P.4)* *Ch. 3.3.3.2 (P.55)*
Compare AI models to identify best performer	*Conducted comparison of several AI models and ANN*	*Ch. 1.2 (P.4)* *Ch. 3.3.3.2 (P.55)*

	was determined to be the best performer.	
Clarification in method for testing result validity	*It involves comparing the system's output with actual laboratory test results to verify accuracy, consistency, and reliability, as discussed in the Data Collection and Testing Plan and Procedures sections.*	*Ch. 3.2 (P.41, 42)*
Correct flowchart symbols and flow	*Corrected overall flowchart design, including proper symbols and flow direction.*	*Ch. 3.3.2(P.45,46)* *Ch. 3.3.2.1(P.48, 50)* *Ch. 3.3.2.2(P.51)* *Ch. 3.3.2.3(P.53)*
Add missing image labels	*Added figure labels and corresponding descriptions.*	*Ch. 3.3.3.3(P.57, 58, 59)*
Change TA name on approval sheet to black	*Changed red text color to black.*	*p. II*
Improve app design for result readability	*Presented screenshots individually for clarity.*	*Ch. 3.3.3.3(P.60, 61, 62, 63)*
Specific objective revision	*Revised objectives by removing specific AI model algorithms and preprocessing techniques to allow flexibility in exploring various combinations for achieving the best accuracy.*	*Ch. 1.2 (P.4)*
Adding Hand Texture in RRL	*Added literature and studies related to hand texture (Normal and High).*	*Ch. 2.1.6 (P. 18, 19, 20)* *Ch. 2.2.6 (P. 31, 32, 33, 34, 35)*
Adding Image Preprocessing in RRL	*Added literature and studies related to image preprocessing.*	*Ch. 2.1.7 (P. 20, 21)* *Ch. 2.2.7 (P. 35, 36, 37)*
Missing discussion of the Testing Plans and algorithm performance comparison	*Added a table about Algorithm Comparative Performance Table showing results of various preprocessing methods and AI models with performance metrics and discussion.*	*Ch. 3.4 (P. 67, 68)*
Include a procedure on how to use the device.	*Added Section for the user's guidelines with a clear step-by-step guide on how to use the device.*	*Ch. 3.3.3.4 (P.64, 65)*
List of tables should be in Title Case and Regular Font	*The List of Tables was updated from all caps to*	*p. IV*

	title case with a regular font for improved readability and consistency.	
Consistency of the word "*study*" in the entire paper	*All instances of "Research" were changed to "Study" to maintain consistent terminology throughout the paper.*	*Entire Paper*
Missing software methodology section	*A dedicated Software Methodology section was added under Chapter 3 to describe the approach used in developing the software.*	*Ch. 3.3.4 (P.65)*
Add a discussion about the Software Design Development model	*A discussion on the chosen software development model was included to explain the framework guiding the project.*	*Ch. 3.3.4 (P.65)*
Add Classification Model Discussion	*The Classification Model section was added to describe the algorithm, its structure, and its role in the project.*	*Ch. 3.3.4.1 (P.65)*
Add Software tool/platform discussion	*A discussion on the software tools used was added to justify the selection of technologies for development.*	*Ch. 3.3.4.2 (P.66)*
Add discussion for algorithm and processes	*The algorithm section now explains the preprocessing, model, and steps involved in the system's operations.*	*Ch. 3.3.4.3 (P.68)*
Add discussion on each testing plan and procedures table	*A discussion on the testing plan was included.*	*Ch. 3.4 (P.69, 70, 71, 72, 73, 74)*
Add discussion on AI model performance metrics. Discuss and add necessary formulas with discussion and tables	*Added a discussion of AI model performance metrics.*	*Ch. 3.4 (P.73,74)*
Add comparison of model performance vs previous studies	*A comparison table and discussion were added to test the model's performance against previous studies.*	*Ch. 3.4 (P.73,74)*
Revise comparative table of the testing plans and algorithm performance comparison	*The comparative table was revised for clarity, showing differences in testing strategies and algorithm effectiveness.*	*Ch. 3.4 (P.72,73)*

C Complied NC Not Complied	Examination Committee	Signature	Date
Complied	ENGR. ROSALINA ESTACIO		11-06-25
Complied	ENGR. AMOR ELLARES		11-07-2025
Complied	ENGR. JOSHUA GULMATICO		Nov. 16, 2025

REVIEWED BY:

PROGRAM CHAIR : ENGR. ROSALINA ESTACIO DATE : 11-06-25

DEAN : ENGR. LIZA R. REYES DATE : _____